Open API

Public information

This article is the open api document of Fox ESS Cloud Platform. Before using this api, please go to the api management of the cloud platform personal center to generate your api-key.

Please make sure to keep your API key safe to avoid any unauthorized access. If it gets lost or leaked, you can go to the personal center to get a new one, and the old key will no longer be valid.

The open API is designed in RESTful style, and the returned data are all in json format, and the encoding format is utf-8.

If you use script calling, be sure to modify your User-Agent.

Change Log

Updated date	Version	Update content
2023/11/30	V1.0.1	The real-time data and historical data interfaces add the sn parameter of the device that can be specified.
2023/12/18	V1.0.2	Added an interface for obtaining variable names. The real-time data and historical data interfaces cancel the limit on the number of variables, and can obtain all variable data by default without passing variables.
2024/01/12	V1.0.3	Added inverter power generation, inverter reports, inverter battery SOC settings, inverter battery strong charging time settings, inverter list, inverter details, collector list, power station list, power station details interface
2024/06/13	V1.0.4	Added battery throughput variables, which can be queried through the real-time data interface
2024/08/19	V1.0.5	Real-time data interface adds V1 version, supports batch query
2024/09/13	V1.0.6	The device parameter setting interface is added, which can now support ExoprtLimit setting
2024/10/18	V1.0.7	Device parameter settings now support H1 series
2024/11/27	V1.0.8	Added equipment time modification function and safety regulation modification function
2024/12/28	V1.0.9	Added the SOH variable, and add the PV power generation data in the report
2025/02/13	V1.0.10	Scheduler interface adds V1 version, supports the maximum soc value set and get
2025/02/18	V1.0.11	Add battery information to device details
2025/03/20	V1.1.0	Add Oauth 2.0 Support
2025/04/10	V1.1.1	Support for some new models
2025/05/12	V1.1.2	Add meter support, add PeakShaving settings
2025/06/03	V1.1.3	Add battery model and capacity to the device details interface, and add work mode parameters to the device settings interface
2025/07/24	V1.1.4	Add micro-reverse and micro-storage related function interfaces
2025/08/13	V1.1.5	Add interface for obtaining communication device signal strength
2025/10/21	V1.1.6	Added battery heating functionality, Modbus transparent transmission capability, and V2 scheduler functionality
2025/10/31	V1.1.7	Added MicroStorage LCD sleep mode functionality, MicroStorage ecomode functionality, and added EMS support
2025/11/28	V1.1.8	Enhance gmax-related functionality and supplement/refine certain interface documentation
2026/01/09	V1.1.9	Added additional device configuration options, including AI Link/EMS-related settings, and added collector version information
2026/02/06	V1.1.10	Add real-time query interface and historical query interface for EMS devices. Add Scheduler V3 version interface.
2026/03/09	V1.1.11	Add boarding status and Data Logger Details endpoint
2026/03/20	v1.1.12	Device param setting APIs echo input on success; gmax peak time params now constrained; third-party PV indicator added to the device detail endpoint.
2026/04/02	v1.1.13	Add device fault history endpoint; Add variable "PV energy total" to real-time data endpoint; Scheduler interface now supports H3Plus.
2026/04/03	v1.1.14	**Add Kafka integration documentation**
2026/04/29	v1.1.15	Add device boarding status retrieve endpoint and support new data variables.
2026/05/07	v1.1.16	**Update OAuth guide**
2026/05/22	v1.1.17	Add battery information to plant details endpoint (US only)
2026/05/29	v1.1.18	Add heat pump endpoints and update variable table.

Request domain https://www.foxesscloud.com/

Request header requirements

Params	Required or not	Example	Parameter Description
token	No		The api-key generated by the API management function in the platform personal center
timestamp	Yes		Current timestamp milliseconds
signature	Yes		Encrypt the string url+"\r\n"+token+"\r\n"+timestamp with md5
lang	Yes	en	langurage
Authorization	No	Bearer **********	Oauth2 authorized accesstoken prefix: Bearer

We support two types of authentication, private tokens and Oauth tokens. If you choose private token, please use token in header, if you choose Oauth token, please use Authorization in header, you can't use both at the same time.

OAuth Guide

This part will walk you through the complete process, from creating your application client to managing device binding. We currently support two authorization grant types:

- **Authorization Code Grant**: For apps accessing resources **on behalf of a user** (e.g., web/mobile apps).
- **Client Credentials Grant**: For **server-to-server** communication (e.g., backend services).

Once you have a valid `access_token`, you can use our APIs to onboard, check, update, and offboard devices.

Authorization Grant Type Availability

We currently offer different grant types depending on your use case:

- **For individual users and non-VPP applications:** We currently support **only the Authorization Code Grant**. This is the standard flow for third-party applications that need to access user-owned device data.
- **For Virtual Power Plant (VPP) operators:** To use either the Authorization Code Grant or Client Credentials Grant, you must first create your client application in our **v1 Platform** and **contact us with your `clientId`**. VPP-specific features will be enabled for your client in our system. Once completed, you can proceed with the OAuth flows described below.

Step 0: Create Your Client Application (Prerequisite)

Before using any OAuth flow, you **must first register your application** in our platform.

How to Create a Client

1. Log in to our **v1 Platform**.
2. Click your **avatar** in the top-right corner.
3. Select **"User Profile"**.
4. In the left navigation menu, go to **"API Management"** → **"Authorization Token"**.

5. Click **"Create Client Side"**.
6. Fill in the form as shown below:
    - **Client Name**: A descriptive name for your application (e.g., `MyVPPApp`).
    - **Redirect Address**: The exact callback URL where users will be redirected after authorization. *(This becomes your `redirect_uri`)*
    - **Scopes**: Select the permissions your app needs:
        - `data_access`: Grants read access to device data.
        - `device_control`: Grants control permissions for devices.
        - *(These selections become your `scope` parameter)*
    - Click **"Save"**.

Critical Notes

- After creation, the system will display your **`client_id`** and **`client_secret`**.
- **The `client_secret` is shown only once.** Please copy and store it securely.
- If you lose the `client_secret`, we recommend you **delete the old client and create a new one**.

Mapping to API Parameters:

- *`client_id` in the API = The `Client ID` generated on this page.*
- *`client_secret` in the API = The `Client Secret` shown after creation.*
- *`redirect_uri` in the API = The `Redirect Address` you entered.*
- *`scope` in the API = The scopes you selected (e.g., `data_access,device_control`).*

1. Authorization Code Grant (On Behalf of a User)

In this flow, the **end-user** is the device owner, and your application must first obtain their explicit consent.

Step 1: Redirect User for Authorization

Redirect the user to our authorization page. The user will log in and grant consent.

URL Example:

```
https://{domain}/h5/auth/foxessIndex?
    response_type=code&
    client_id={your_client_id}&
    redirect_uri={your_redirect_uri}&
    scope={your_scopes}
```

Step 2: Receive the Authorization Code

After the user grants consent, the browser is redirected back to your `redirect_uri` with a temporary `code` appended to the URL.

Callback Example:

```
{your_redirect_uri}?code={authorization_code}&deviceSn={device_serial_number}
```

Step 3: Exchange Code for Access Token

Your application server must immediately exchange this `code` for a long-lived `access_token` and a `refresh_token`.

Request Example:

```
POST https://{domain}/oauth2/token
Content-Type: application/x-www-form-urlencoded
  grant_type=authorization_code&

  client_id={your_client_id}&

  client_secret={your_client_secret}&

  code=ABC123&

  redirect_uri={your_redirect_uri}
```

Note: The `access_token` is valid for 24 hours. Use the `refresh_token` to obtain a new one before it expires.

Step 4: Device Onboarding (Automatic)

During the **Step 1** authorization flow, our system will **automatically onboard all devices under the user's account (or the user-selected devices) to your `client_id`** upon their consent.

Step 5: Check Device Boarding Status (Optional)

Using the `access_token` from Step 3, you can query the current boarding status of a specific device at any time.

Request Example:

```
GET https://{api_domain}/op/v0/device/boarding/status?sn=YOUR_DEVICE_SN
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
```

Step 6: Update Device Boarding Status (Optional)

You can update a device's boarding status based on your business logic (e.g., changing from `0` to `1`).

Request Example:

```
POST https://{api_domain}/op/v0/device/boarding/status/set
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
Content-Type: application/json
{
    "sn": "YOUR_DEVICE_SN",
    "status": 1
}
```

Step 7: Offboard Devices (Optional)

When a user wishes to revoke authorization, you can offboard devices using the following methods.

- **Revoke Single Token**:

```
GET https://{domain}/oauth2/revok?client_id={your_client_id}&client_secret={your_client_secret}&access_token={your_access_token}
```

- **Revoke All Tokens**:

```
GET https://{domain}/oauth2/revokeAllTokens?client_id={your_client_id}&client_secret={your_client_secret}&access_token={your_access_token}
```

Step 8: Refresh Access Token

When the `access_token` expires (after 24 hours) or is about to expire, use the `refresh_token` to get a new one.

Request Example:

```
POST https://{domain}/oauth2/refresh
Content-Type: application/x-www-form-urlencoded
  grant_type=refresh_token&

  client_id={your_client_id}&

  client_secret={your_client_secret}&

  refresh_token={your_refresh_token}
```

---

2. Client Credentials Grant (For Your Own Service)

In this flow, **your application itself** is the owner or manager of the devices, and no user interaction is required.

Step 1: Obtain an Access Token Directly

Your application server uses its `client_id` and `client_secret` to request an `access_token` directly.

Request Example:

```
POST https://{domain}/oauth2/client_token
Content-Type: application/x-www-form-urlencoded
  grant_type=client_credentials&

  client_id={your_client_id}&

  client_secret={your_client_secret}
```

> *Note: This grant type does not provide a `refresh_token`. To get a new token after expiration, simply repeat this request.*

Step 2: Onboard Devices Manually

Unlike the Authorization Code flow, **devices are NOT onboarded automatically** in this mode. You must explicitly call an API to bind a device.

Request Example:

```
POST https://{api_domain}/op/v0/vpp/oauth2/client/onboard
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
Content-Type: application/json
{
    "deviceSN": "YOUR_DEVICE_SN"
}
```

Step 3: Check and Update Device Boarding Status (Optional)

The logic is identical to the Authorization Code flow.

Check Status:

```
GET https://{api_domain}/op/v0/device/boarding/status?sn=YOUR_DEVICE_SERIAL_NUMBER
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
```

Update Status:

```
POST https://{api_domain}/op/v0/device/boarding/status/set
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
Content-Type: application/json
{
    "sn": "YOUR_DEVICE_SN",
    "status": 1
}
```

Step 4: Offboard Devices (Optional)

Request Example:

```
POST https://{api_domain}/op/v0/vpp/oauth2/client/offboard
Authorization: Bearer YOUR_ACCESS_TOKEN
Signature: Encrypt the string url+"\r\n"+ YOUR_ACCESS_TOKEN +"\r\n"+timestamp with md5
Content-Type: application/json
{
    "deviceSN": "DEVICE_SN"
}
```

Step 5: Refresh Access Token

After the token expires, simply **repeat Step 1** to obtain a new `access_token`.

Access frequency limit

- Each inverter under a single account has a total of 1440 interface calls per day.
- The query interface is limited to once per second, and each interface is calculated separately.
- Inserting update interfaces is limited to once every 2 seconds, and each interface is calculated separately.

This limit is calculated based on various factors such as server performance, and the Company reserves the right of final interpretation

Common error solutions

40256 The request header parameters are missing. Please check whether the request headers are consistent with the document requirements.

40257 The request body parameters are invalid. Please check whether the request body is consistent with the document requirements.

40400 The number of requests is too frequent. Please reduce the frequency of access.

Code example

```
import json
import time
import urllib3
import requests
import hashlib
import os
from json import JSONDecodeError
 """debug mode"""

 debug = False

 """request interval (second)"""

 sleep_time = 0

 """domain name, do not modify it unless necessary"""

 domain = 'https://portal.foxesscloud.us'

 """your key"""

 key = ''

 class GetAuth:

 def get_signature(self, token, path, lang='en'):

 """

 This function is used to generate a signature consisting of URL, token, and timestamp, and return a dictionary containing the signature and other information.

 :param token: your key

 :param path:  your request path

 :param lang: language, default is English.

 :return: with authentication header

 """

 timestamp = round(time.time() * 1000)

 signature = fr'{path}\r\n{token}\r\n{timestamp}'
```

```
 result = {
     'token': token,
     'lang': lang,
     'timestamp': str(timestamp),
     'signature': self.md5c(text=signature),
     'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) '
                   'Chrome/117.0.0.0 Safari/537.36'
 }
 return result
```

```
@staticmethod

                def md5c(text="", _type="lower"):

                res = hashlib.md5(text.encode(encoding='UTF-8')).hexdigest()

                if _type.eq("lower"):

                return res

                else:

                return res.upper()


def save_response_data(response, filename):


"""Create the 'data' directory if it doesn't exist"""


os.makedirs('data', exist_ok=True)


"""Concatenate the directory path and filename to create the full file path"""

                file_path = os.path.join('data', filename)


"""Get the current timestamp"""

                timestamp = time.strftime("%Y-%m-%d %H:%M:%S", time.localtime())


"""Check if the status code is 200"""

                if response.status_code == 200:

                """Save the response data and timestamp to a dictionary"""

                try:

                response_json = response.json()

                except JSONDecodeError:

                response_json = response.text
```

```
data = {
    'response': response_json,
    'timestamp': timestamp
}
"""Write the dictionary as JSON to the file"""
with open(file_path, 'w', encoding='utf-8') as file:
    json.dump(data, file, ensure_ascii=False, indent=4)
```

```
else:
                                    """Save the status code, response text, and timestamp to a dictionary"""
                                    data = {
                                    'status_code': response.status_code,
                                    'response_text': response.text,
                                    'timestamp': timestamp
                                    }
                                    """Write the dictionary as JSON to the file"""
                                    with open(file_path, 'w', encoding='utf-8') as file:
                                    json.dump(data, file, ensure_ascii=False, indent=4)

print(f'{timestamp}: [{filename}] successfully saved')


urllib3.disable_warnings()


def fr_requests(method, path, param=None):


url = domain + path


headers = GetAuth().get_signature(token=key, path=path)


time.sleep(sleep_time)


if method == 'get':
                                    response = requests.get(url=url, params=param, headers=headers, verify=False)


elif method == 'post':
                                    response = requests.post(url=url, json=param, headers=headers, verify=False)
                                    else:
                                    raise Exception('request method error')


if debug:
                                    result = {'url': url, 'method': method, 'param': param, 'headers': headers, 'response': response.text}
                                    print(json.dumps(result, indent=1))
                                    print('-------------------------' * 5)
```

```
                                return response


class Plant:


@staticmethod
                                def plant_list():
                                path = '/op/v0/plant/list'
                                request_param = {'currentPage': 1, 'pageSize': 10}
                                response = fr_requests('post', path, request_param)
                                save_response_data(response, 'plant_list_response.json')
                                return response


@staticmethod
                                def plant_detail():
                                path = '/op/v0/plant/detail'
                                request_param = {'id': 'abc'}
                                response = fr_requests('get', path, request_param)
                                save_response_data(response, 'plant_detail_response.json')
                                return response


class Device:


@staticmethod


def device_list():


path = '/op/v0/device/list'


request_param = {'currentPage': 1, 'pageSize': 500}


response = fr_requests('post', path, request_param)


save_response_data(response, 'device_list_response.json')


return response
```

```
@staticmethod
                                    def device_detail():
                                    path = '/op/v0/device/detail'
                                    request_param = {'sn': 'sn'}
                                    response = fr_requests('get', path, request_param)
                                    save_response_data(response, 'device_detail_response.json')
                                    return response

@staticmethod
                                    def variable_get():
                                    path = '/op/v0/device/variable/get'
                                    response = fr_requests('get', path)
                                    save_response_data(response, 'device_variable_get_response.json')
                                    return response

@staticmethod
                                    def real_query():
                                    path = '/op/v0/device/real/query'
                                    request_param = {'sn': 'sn', 'variables': []}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_real_query_response.json')
                                    return response

@staticmethod
                                    def history_query():
                                    path = '/op/v0/device/history/query'
                                    """get the current millisecond level timestamp"""
                                    end_time = int(time.time() * 1000)
                                    """timestamp 24 hours ago"""
                                    begin_time = end_time - 3600000
                                    request_param = {'sn': 'sn', 'variables': [], 'begin': begin_time, 'end': end_time}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_history_query_response.json')
                                    return response
```

```
@staticmethod
                                    def report_query():
                                    path = '/op/v0/device/report/query'
                                    request_param = {"sn": "sn",
                                    "year": 2024, "month": 1, 'day': 17, "dimension": "day",
                                    "variables": ["generation", "feedin", "gridConsumption",
                                    "chargeEnergyToTal", "dischargeEnergyToTal"]}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_report_query_response.json')
                                    return response

@staticmethod
                                    def device_generation():
                                    path = '/op/v0/device/generation'
                                    request_param = {'sn': 'sn'}
                                    response = fr_requests('get', path, request_param)
                                    save_response_data(response, 'device_generation_response.json')
                                    return response

@staticmethod
                                    def device_bat_soc_get():
                                    path = '/op/v0/device/battery/soc/get'
                                    request_param = {'sn': 'sn'}
                                    response = fr_requests('get', path, request_param)
                                    save_response_data(response, 'device_bat_soc_get_response.json')
                                    return response

@staticmethod
                                    def device_bat_soc_set():
                                    path = '/op/v0/device/battery/soc/set'
                                    request_param = {'sn': 'sn', 'minSoc': 10, 'minSocOnGrid': 10}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_bat_soc_set_response.json')
```

```
                                return response


@staticmethod
                                def device_bat_force_charge_time_get():
                                path = '/op/v0/device/battery/forceChargeTime/get'
                                request_param = {'sn': 'sn'}
                                response = fr_requests('get', path, request_param)
                                save_response_data(response, 'device_bat_force_charge_time_get_response.json')
                                return response


@staticmethod
                                def device_bat_force_charge_time_set():
                                path = '/op/v0/device/battery/forceChargeTime/set'
                                request_param = {"sn": "sn", "enable1": True, "enable2": False,
                                "startTime1": {"hour": 12, "minute": 0}, "endTime1": {"hour": 14, "minute": 56},
                                "startTime2": {"hour": 2, "minute": 0}, "endTime2": {"hour": 4, "minute": 0}}
                                """
                                request_param = {"sn": "sn", "enable1": False, "enable2": False,
                                "startTime1": {"hour": 11, "minute": 1}, "endTime1": {"hour": 15, "minute": 57},
                                "startTime2": {"hour": 3, "minute": 1}, "endTime2": {"hour": 5, "minute": 1}}
                                """
                                response = fr_requests('post', path, request_param)
                                save_response_data(response, 'device_bat_force_charge_time_set_response.json')
                                return response


@staticmethod
                                def device_scheduler_get_flag():
                                path = '/op/v0/device/scheduler/get/flag'
                                request_param = {'deviceSN': 'sn'}
                                response = fr_requests('post', path, request_param)
                                save_response_data(response, 'device_scheduler_get_flag_response.json')
                                return response


@staticmethod
```

```
                                    def device_scheduler_set_flag():
                                    path = '/op/v0/device/scheduler/set'
                                    request_param = {'deviceSN': 'sn', 'enable': 0}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_scheduler_set_flag_response.json')
                                    return response

@staticmethod
                                    def device_scheduler_get():
                                    path = '/op/v0/device/scheduler/get'
                                    request_param = {'deviceSN': 'sn'}
                                    response = fr_requests('post', path, request_param)
                                    save_response_data(response, 'device_scheduler_get_response.json')
                                    return response

@staticmethod
                                    def device_scheduler_enable():
                                    path = '/op/v0/device/scheduler/enable'
                                    request_param1 = {"deviceSN": "sn",
                                    "groups": [{"enable": 1, "startHour": 0, "startMinute": 0, "endHour": 1, "endMinute": 59,
                                    "workMode": "SelfUse", "minSocOnGrid": 11, "fdSoc": 12, "fdPwr": 5001},
                                    {"enable": 1, "startHour": 2, "startMinute": 1, "endHour": 3, "endMinute": 0,
                                    "workMode": "SelfUse", "minSocOnGrid": 21, "fdSoc": 22, "fdPwr": 5002},
                                    {"enable": 1, "startHour": 3, "startMinute": 1, "endHour": 3, "endMinute": 58,
                                    "workMode": "Feedin", "minSocOnGrid": 31, "fdSoc": 32, "fdPwr": 5003},
                                    {"enable": 1, "startHour": 4, "startMinute": 1, "endHour": 4, "endMinute": 58,
                                    "workMode": "Backup", "minSocOnGrid": 41, "fdSoc": 42, "fdPwr": 5004},
                                    {"enable": 1, "startHour": 5, "startMinute": 1, "endHour": 5, "endMinute": 58,
                                    "workMode": "ForceCharge", "minSocOnGrid": 51, "fdSoc": 52, "fdPwr": 5005},
                                    {"enable": 1, "startHour": 6, "startMinute": 1, "endHour": 6, "endMinute": 58,
                                    "workMode": "ForceDischarge", "minSocOnGrid": 61, "fdSoc": 62, "fdPwr": 5006},
                                    {"enable": 1, "startHour": 7, "startMinute": 0, "endHour": 7, "endMinute": 59,
                                    "workMode": "ForceDischarge", "minSocOnGrid": 71, "fdSoc": 72, "fdPwr": 0},
                                    {"enable": 1, "startHour": 8, "startMinute": 0, "endHour": 23, "endMinute": 59,
```

```
                                    "workMode": "ForceDischarge", "minSocOnGrid": 81, "fdSoc": 82, "fdPwr": 6000}]}
                                    request_param2 = {"deviceSN": "sn",
                                    "groups": [{"enable": 0, "startHour": 0, "startMinute": 0, "endHour": 0, "endMinute": 1,
                                    "workMode": "ForceCharge", "minSocOnGrid": 10, "fdSoc": 0, "fdPwr": 0},
                                    {"enable": 0, "startHour": 3, "startMinute": 2, "endHour": 4, "endMinute": 1,
                                    "workMode": "ForceCharge", "minSocOnGrid": 22, "fdSoc": 23, "fdPwr": 5003},
                                    {"enable": 0, "startHour": 4, "startMinute": 2, "endHour": 4, "endMinute": 59,
                                    "workMode": "Backup", "minSocOnGrid": 32, "fdSoc": 32, "fdPwr": 5004},
                                    {"enable": 0, "startHour": 5, "startMinute": 2, "endHour": 5, "endMinute": 59,
                                    "workMode": "Feedin", "minSocOnGrid": 42, "fdSoc": 42, "fdPwr": 5005},
                                    {"enable": 0, "startHour": 6, "startMinute": 2, "endHour": 6, "endMinute": 59,
                                    "workMode": "SelfUse", "minSocOnGrid": 52, "fdSoc": 52, "fdPwr": 5006},
                                    {"enable": 0, "startHour": 7, "startMinute": 1, "endHour": 7, "endMinute": 59,
                                    "workMode": "SelfUse", "minSocOnGrid": 62, "fdSoc": 62, "fdPwr": 5007},
                                    {"enable": 0, "startHour": 8, "startMinute": 1, "endHour": 8, "endMinute": 3,
                                    "workMode": "SelfUse", "minSocOnGrid": 72, "fdSoc": 72, "fdPwr": 0},
                                    {"enable": 0, "startHour": 22, "startMinute": 59, "endHour": 23, "endMinute": 59,
                                    "workMode": "SelfUse", "minSocOnGrid": 100, "fdSoc": 100, "fdPwr": 6000}]}
                                    response = fr_requests('post', path, request_param2)
                                    save_response_data(response, 'device_scheduler_set_response.json')
                                    return response

class Module:

@staticmethod

def module_list():

path = '/op/v0/module/list'

request_param = {'currentPage': 1, 'pageSize': 10}

response = fr_requests('post', path, request_param)

save_response_data(response, 'module_list_response.json')
```

```
return response

class User:

@staticmethod

def user_get_access_count():

path = '/op/v0/user/getAccessCount'

response = fr_requests('get', path)

save_response_data(response, 'user_get_access_count_response.json')

return response

if name == 'main':

plant = Plant()

"""
```

plant.plant_list()

```
plant.plant_detail()

"""

device = Device()
                                """
                                device.device_list()
                                device.device_detail()
                                device.variable_get()
                                device.real_query()
```

```
                                device.history_query()
                                device.report_query()
                                device.device_generation()

device.device_bat_soc_set()
                                time.sleep(3)
                                device.device_bat_soc_get()

device.device_bat_force_charge_time_set()
                                time.sleep(3)
                                device.device_bat_force_charge_time_get()

device.device_scheduler_set_flag()
                                time.sleep(3)
                                device.device_scheduler_get_flag()
                                device.device_scheduler_enable()
                                time.sleep(3)
                                device.device_scheduler_get()
                                """

module = Module()
                                """
                                module.module_list()
                                """

user = User()
                                """
                                user.user_get_access_count()
                                """
```

Variable table

The table presented below offers a comprehensive overview of the variables that can be accessed through the API. It is important to note that the availability of these variables may differ depending on the specific device being utilized. Therefore, the variables listed in the table are subject to change based on the device in use.

PV (Photovoltaic)

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
todayYield	kW	Energy generated today	TRUE	TRUE
PVEnergyTotal	kWh	Total photovoltaic power generation (PV panel side)	TRUE	TRUE
pvPower	kW	Total PV input power	TRUE	TRUE
pv1Volt	V	PV1 voltage	TRUE	TRUE
pv1Current	A	PV1 current	TRUE	TRUE
pv1Power	kW	PV1 power	TRUE	TRUE
pv2Volt	V	PV2 voltage	TRUE	TRUE
pv2Current	A	PV2 current	TRUE	TRUE
pv2Power	kW	PV2 power	TRUE	TRUE
pv3Volt	V	PV3 voltage	TRUE	TRUE
pv3Current	A	PV3 current	TRUE	TRUE
pv3Power	kW	PV3 power	TRUE	TRUE
pv4Volt	V	PV4 voltage	TRUE	TRUE
pv4Current	A	PV4 current	TRUE	TRUE
pv4Power	kW	PV4 power	TRUE	TRUE
pv5Volt	V	PV5 voltage	TRUE	TRUE
pv5Current	A	PV5 current	TRUE	TRUE
pv5Power	kW	PV5 power	TRUE	TRUE
pv6Volt	V	PV6 voltage	TRUE	TRUE
pv6Current	A	PV6 current	TRUE	TRUE
pv6Power	kW	PV6 power	TRUE	TRUE
pv7Volt	V	PV7 voltage	TRUE	TRUE
pv7Current	A	PV7 current	TRUE	TRUE
pv7Power	kW	PV7 power	TRUE	TRUE
pv8Volt	V	PV8 voltage	TRUE	TRUE
pv8Current	A	PV8 current	TRUE	TRUE
pv8Power	kW	PV8 power	TRUE	TRUE
pv9Volt	V	PV9 voltage	TRUE	TRUE
pv9Current	A	PV9 current	TRUE	TRUE
pv9Power	kW	PV9 power	TRUE	TRUE
pv10Volt	V	PV10 voltage	TRUE	TRUE
pv10Current	A	PV10 current	TRUE	TRUE

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
pv10Power	kW	PV10 power	TRUE	TRUE
pv11Volt	V	PV11 voltage	TRUE	TRUE
pv11Current	A	PV11 current	TRUE	TRUE
pv11Power	kW	PV11 power	TRUE	TRUE
pv12Volt	V	PV12 voltage	TRUE	TRUE
pv12Current	A	PV12 current	TRUE	TRUE
pv12Power	kW	PV12 power	TRUE	TRUE
pv13Volt	V	PV13 voltage	TRUE	TRUE
pv13Current	A	PV13 current	TRUE	TRUE
pv13Power	kW	PV13 power	TRUE	TRUE
pv14Volt	V	PV14 voltage	TRUE	TRUE
pv14Current	A	PV14 current	TRUE	TRUE
pv14Power	kW	PV14 power	TRUE	TRUE
pv15Volt	V	PV15 voltage	TRUE	TRUE
pv15Current	A	PV15 current	TRUE	TRUE
pv15Power	kW	PV15 power	TRUE	TRUE
pv16Volt	V	PV16 voltage	TRUE	TRUE
pv16Current	A	PV16 current	TRUE	TRUE
pv16Power	kW	PV16 power	TRUE	TRUE
pv17Volt	V	PV17 voltage	TRUE	TRUE
pv17Current	A	PV17 current	TRUE	TRUE
pv17Power	kW	PV17 power	TRUE	TRUE
pv18Volt	V	PV18 voltage	TRUE	TRUE
pv18Current	A	PV18 current	TRUE	TRUE
pv18Power	kW	PV18 power	TRUE	TRUE
pv19Volt	V	PV19 voltage	TRUE	TRUE
pv19Current	A	PV19 current	TRUE	TRUE
pv19Power	kW	PV19 power	TRUE	TRUE
pv20Volt	V	PV20 voltage	TRUE	TRUE
pv20Current	A	PV20 current	TRUE	TRUE
pv20Power	kW	PV20 power	TRUE	TRUE
pv21Volt	V	PV21 voltage	TRUE	TRUE

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
pv21Current	A	PV21 current	TRUE	TRUE
pv21Power	kW	PV21 power	TRUE	TRUE
pv22Volt	V	PV22 voltage	TRUE	TRUE
pv22Current	A	PV22 current	TRUE	TRUE
pv22Power	kW	PV22 power	TRUE	TRUE
pv23Volt	V	PV23 voltage	TRUE	TRUE
pv23Current	A	PV23 current	TRUE	TRUE
pv23Power	kW	PV23 power	TRUE	TRUE
pv24Volt	V	PV24 voltage	TRUE	TRUE
pv24Current	A	PV24 current	TRUE	TRUE
pv24Power	kW	PV24 power	TRUE	TRUE

Grid

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
RCurrent	A	Grid R-phase current	TRUE	TRUE
RVolt	V	Grid R-phase voltage	TRUE	TRUE
RFreq	Hz	Grid R-phase frequency	TRUE	TRUE
RPower	kW	Grid R-phase active power	TRUE	TRUE
SCurrent	A	Grid S-phase current	TRUE	TRUE
SVolt	V	Grid S-phase voltage	TRUE	TRUE
SFreq	Hz	Grid S-phase frequency	TRUE	TRUE
SPower	kW	Grid S-phase active power	TRUE	TRUE
TCurrent	A	Grid T-phase current	TRUE	TRUE
TVolt	V	Grid T-phase voltage	TRUE	TRUE
TFreq	Hz	Grid T-phase frequency	TRUE	TRUE
TPower	kW	Grid T-phase active power	TRUE	TRUE
feedinPower	kW	Power exported to grid	TRUE	TRUE
gridConsumptionPower	kW	Power drawn from grid	TRUE	TRUE
ReactivePower	kVar	Reactive power	TRUE	TRUE
PowerFactor		Power factor	TRUE	TRUE
generation	kWh	Cumulative power generation (AC output from inverter side, affected by battery charging/discharging	TRUE	TRUE
gridConsumption	kWh	Total grid electricity consumption	TRUE	TRUE
gridConsumption2	kWh	Total electricity consumption of Meter 2	TRUE	TRUE
gridConsumption3	kWh	Total electricity consumption of Meter 3	TRUE	TRUE
feedin	kWh	Total feed-in energy	TRUE	TRUE
feedin2	kWh	Total feed-in energy of Meter 2	TRUE	TRUE

Loads

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
loads	kWh	Load power consumption	TRUE	TRUE
loadsPower	kW	Total load power	TRUE	TRUE
loadsPowerR	kW	Load R-phase power	TRUE	TRUE
loadsPowerS	kW	Load S-phase power	TRUE	TRUE
loadsPowerT	kW	Load T-phase power	TRUE	TRUE

Inverter

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
ambientTemperation	°C	Ambient temperature	TRUE	TRUE
boostTemperation	°C	Boost circuit temperature	TRUE	TRUE
invTemperation	°C	Inverter temperature	TRUE	TRUE
chargeTemperature	°C	Charging module temperature	TRUE	TRUE
dspTemperature	°C	DSP chip temperature	TRUE	TRUE
generationPower	kW	Total AC output power	TRUE	TRUE
runningState		Running state	TRUE	TRUE
currentFault		Current fault code	TRUE	TRUE
currentFaultCount		Current fault count	TRUE	TRUE

Battery

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
SoC	%	State of Charge	TRUE	FALSE
SOH	%	State of Health	TRUE	FALSE
batStatus		Battery status code	TRUE	FALSE
batStatusV2		Battery status name	TRUE	FALSE
batTemperature	°C	Battery temperature	TRUE	FALSE
batVolt	V	Battery voltage	TRUE	FALSE
batCurrent	A	Battery current	TRUE	FALSE
batChargePower	kW	Battery charge powerNote: check note for "invBatPower"	TRUE	FALSE
batDischargePower	kW	Battery discharge powerNote: check note for "invBatPower"	TRUE	FALSE
invBatVolt	V	Inverter-side battery voltage	TRUE	TRUE
invBatCurrent	A	Inverter-side battery currentNote: Positive Discharge, Negative Charge	TRUE	FALSE
invBatPower	kW	Inverter-side battery powerNote: Positive Discharge, Negative Charge;is split into two separate positive values:"batDischargePower" (when positive)and"batChargePower" (absolute value when negative).	TRUE	FALSE
maxChargeCurrent	A	Maximum charge current	TRUE	FALSE
maxDischargeCurrent	A	Maximum discharge current	TRUE	FALSE
remainCapacity	Ah	Remaining capacity	TRUE	FALSE
RemainingPowerCapability	Ah	Remaining power capability	TRUE	FALSE
ResidualEnergy	kWh	Remaining energy in battery	TRUE	FALSE
batCycleCount		Number of charge/discharge cycles	TRUE	FALSE
totalDischargeKW	kWh	Battery total discharge energy	TRUE	FALSE
totalDischargeAh	Ah	Battery total discharge capacity	TRUE	FALSE
chargeEnergyToTal	kWh	Total charge energy	TRUE	FALSE
dischargeEnergyToTal	kWh	Total discharge energy	TRUE	FALSE
energyThroughput	kWh	Total energy cycled (lifetime)	TRUE	FALSE
EventNumber		Adverse event count	TRUE	FALSE

Meter

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
meterPower	kW	Meter 1 total active power	TRUE	TRUE
meterPower2	kW	Meter 2 total active power	TRUE	TRUE
meterPowerR	kW	Meter 1 R-phase active power	TRUE	TRUE
meterPowerS	kW	Meter 1 S-phase active power	TRUE	TRUE
meterPowerT	kW	Meter 1 T-phase active power	TRUE	TRUE
meter3Status		Operational status of Meter 3	TRUE	TRUE

EPS (Emergency Power Supply)

Variable Key	Unit	Business Meaning	For Energy-storage	For Grid-tied
epsPower	kW	EPS total output power	TRUE	FALSE
epsCurrentR	A	EPS R-phase current	TRUE	FALSE
epsVoltR	V	EPS R-phase voltage	TRUE	FALSE
epsPowerR	kW	EPS R-phase power	TRUE	FALSE
epsCurrentS	A	EPS S-phase current	TRUE	FALSE
epsVoltS	V	EPS S-phase voltage	TRUE	FALSE
epsPowerS	kW	EPS S-phase power	TRUE	FALSE
epsCurrentT	A	EPS T-phase current	TRUE	FALSE
epsVoltT	V	EPS T-phase voltage	TRUE	FALSE
epsPowerT	kW	EPS T-phase power	TRUE	FALSE

Heat Pump　For Kafka

Variable Key	Business Meaning	Category
indoorDeviceType	Model Name	Heat PumpHardware Identification (Static Data)
capacity	Power Class	Heat PumpHardware Identification (Static Data)
sn	Outdoor Unit S/N	Heat PumpHardware Identification (Static Data)
indoorSN	Indoor Unit S/N & Gateway S/N	Heat PumpHardware Identification (Static Data)
systemFunction	Domestic Hot Water installed	Device Details(Semi Static Data)
Z1ZoneSwitch	Heating Circuits	Device Details(Semi Static Data)
Z2ZoneSwitch	Heating Circuits	Device Details(Semi Static Data)
Z3ZoneSwitch	Heating Circuits	Device Details(Semi Static Data)
Z1ZoneEmitterType	Heating Circuit Type	Device Details(Semi Static Data)
Z1ZoneMaxWaterTemp	Heating Circuit Type	Device Details(Semi Static Data)
Z2ZoneEmitterType	Heating Circuit Type	Device Details(Semi Static Data)
Z2ZoneMaxWaterTemp	Heating Circuit Type	Device Details(Semi Static Data)
Z3ZoneEmitterType	Heating Circuit Type	Device Details(Semi Static Data)
Z3ZoneMaxWaterTemp	Heating Circuit Type	Device Details(Semi Static Data)
currentRunMode	System Status	Status (Dynamicon Change)
heatPumpConnectivityStatus	Connectivity Status	Status (Dynamicon Change)
sgReadyEnable	Smart Grid Status	Status (Dynamicon Change)
sgReadyMode	§14a status	Status (Dynamicon Change)
sgTclmin	§14a dimming target	Status (Dynamicon Change)
sgThtmax	§14a dimming target	Status (Dynamicon Change)
sgTwmax	§14a dimming target	Status (Dynamicon Change)
outdoorAirTemp	Outdoor temperatureused for weather curve?	Metrics(Dynamic Time series)
dcWaterPumpOFFFlowRate	Hydraulics station flow speed	Metrics(Dynamic Time series)
outletWaterTemp	Hydraulic station flow temperature	Metrics(Dynamic Time series)
returnWaterTemp	Hydraulic station return temperature	Metrics(Dynamic Time series)
Z1ZoneTemp	Flow temperature of each heating circuit	Metrics(Dynamic Time series)
Z2ZoneTemp	Flow temperature of each heating circuit	Metrics(Dynamic Time series)
Z3ZoneTemp	Flow temperature of each heating circuit	Metrics(Dynamic Time series)
dcFanOFFSpeed	Ventilator RPM	Metrics(Dynamic Time series)
dcFanMaxSpeed	Ventilator max RPM	Metrics(Dynamic Time series)
bufferTankTemp	Buffer tank temperature	Metrics(Dynamic Time series)
SolenoidThree-wayValveStatus	3-way-valve position	Metrics(Dynamic Time series)

Variable Key	Business Meaning	Category
flowPressure	Technical Water Pressure	Metrics(Dynamic Time series)
waterTankTemp	Domestic hot water actual temperature	Metrics(Dynamic Time series)
heatingSetTemp	Domestic hot water target temperature	Metrics(Dynamic Time series)
heatingRealTimeConsumption	Power consumption heating	Metrics(Dynamic Time series)
dhwRealTimeConsumption	Power consumption heating domestic hot water	Metrics(Dynamic Time series)
totalIndoorConsumption	Power consumption heating rod heating	Metrics(Dynamic Time series)
heatingRealTimeProduction	Generated heat heating	Metrics(Dynamic Time series)
dhwRealTimeProduction	Generated heat domestic hot water	Metrics(Dynamic Time series)
currentFault	Error Code	Errors & Warnings

Appendix For Enum Variable

Variable Type	Variable Key	Value	Meaning
Inverter	runningState	160	self-test
Inverter	runningState	161	waiting
Inverter	runningState	162	checking
Inverter	runningState	163	on-grid
Inverter	runningState	164	off-grid
Inverter	runningState	165	fault
Inverter	runningState	166	permanent-fault
Inverter	runningState	167	standby
Inverter	runningState	168	upgrading
Inverter	runningState	169	fct
Inverter	runningState	170	illegal

Kafka Integration Documentation

---

1. Overview

This document provides guidance for customers to integrate with our Kafka messaging service.

After completing the integration, customers can use the Kafka protocol to:

- Subscribe to data

---

2. Integration Process

2.1 Integration Steps

The customer integration workflow is as follows:

1. Submit an integration request (provide company/system information)
2. We create a **Kafka non-production environment**:
    - Account (Username / Password)
    - Topic and Group ID
    - Permissions (Read)
3. We provide the integration details
4. Customer completes SDK/client integration
5. Joint debugging and testing
6. Create Kafka production environment
7. Release to production

---

3. Integration Information Description

We will provide the following information:

Parameter	Example	Description
Bootstrap Servers	xxx.kafka.aliyuncs.com:9094	Kafka endpoint
Username	user_xxx	SASL username
Password	******	SASL password
Topic	device-data-prod	Topic name
Consumer Group	group_xxx	Consumer group (optional)
Security Protocol	SASL_SSL	Connection protocol

---

4. Client Integration Examples

4.1 Java Example

```
# kafka.properties
## ============================== Common Config ==============================
bootstrap.servers=xxxxxxxxxxxxxxxxxxxx
topic=xxx
group.id=xxx
 ======================= Configure Based on Your Setup =====================

 SSL endpoint config

 ssl.truststore.location=/xxxx/only.4096.client.truststore.jks

 Must be KafkaOnsClient, do not modify

 ssl.truststore.password=KafkaOnsClient

 Hostname verification (leave empty)

 ssl.endpoint.identification.algorithm=

 java.security.auth.login.config=/xxxx/kafka_client_jaas.conf

 SASL PLAIN

 java.security.auth.login.config.plain=/xxxx/kafka_client_jaas_plain.conf

 SASL SCRAM

 java.security.auth.login.config.scram=/xxxx/kafka_client_jaas_scram.conf
```

Java Kafka Consumer Example (with Detailed Comments)

```
import java.util.ArrayList;
import java.util.List;
import java.util.Properties;

import org.apache.kafka.clients.consumer.ConsumerConfig;
import org.apache.kafka.clients.consumer.ConsumerRecord;
import org.apache.kafka.clients.consumer.ConsumerRecords;
import org.apache.kafka.clients.consumer.KafkaConsumer;
import org.apache.kafka.clients.producer.ProducerConfig;

/*
 * For SSL endpoint instances, uncomment the following three lines.
 * For SASL endpoint instances, uncomment the first two lines below.
 */
// import org.apache.kafka.clients.CommonClientConfigs;
// import org.apache.kafka.common.config.SaslConfigs;
// import org.apache.kafka.common.config.SslConfigs;

public class KafkaConsumerDemo {

    public static void main(String args[]) {

        // Set the JAAS configuration file path
        /*
         * For SSL endpoint instances, uncomment the following line:
         * JavaKafkaConfigurer.configureSasl();
         */

        /*
         * For SASL PLAIN mechanism instances, uncomment the following line:
         * JavaKafkaConfigurer.configureSaslPlain();
         */

        /*
         * For SASL SCRAM mechanism instances, uncomment the following line:
         * JavaKafkaConfigurer.configureSaslScram();
         */

        // Load kafka.properties
        Properties kafkaProperties = JavaKafkaConfigurer.getKafkaProperties();

        Properties props = new Properties();

        // Set bootstrap servers (Kafka endpoint)
        props.put(ProducerConfig.BOOTSTRAP_SERVERS_CONFIG,
                kafkaProperties.getProperty("bootstrap.servers"));

        // Default session timeout (30 seconds)
        props.put(ConsumerConfig.SESSION_TIMEOUT_MS_CONFIG, 30000);

        /*
         * For SSL endpoint instances, uncomment the following configurations:
         *
         * Set SSL truststore path (replace XXX with your actual path)
         * props.put(SslConfigs.SSL_TRUSTSTORE_LOCATION_CONFIG,
         *         kafkaProperties.getProperty("ssl.truststore.location"));
         *
         * Truststore password (must remain unchanged)
         * props.put(SslConfigs.SSL_TRUSTSTORE_PASSWORD_CONFIG, "KafkaOnsClient");
         *
         * Security protocol
         * props.put(CommonClientConfigs.SECURITY_PROTOCOL_CONFIG, "SASL_SSL");
         *
         * SASL mechanism
```

```
         * props.put(SaslConfigs.SASL_MECHANISM, "PLAIN");
         *
         * Max interval between polls
         * props.put(ConsumerConfig.SESSION_TIMEOUT_MS_CONFIG, 30000);
         *
         * Fetch size configuration (important for public network access)
         * props.put(ConsumerConfig.MAX_PARTITION_FETCH_BYTES_CONFIG, 32000);
         * props.put(ConsumerConfig.FETCH_MAX_BYTES_CONFIG, 32000);
         */

        /*
         * For SASL PLAIN mechanism instances, uncomment:
         *
         * props.put(CommonClientConfigs.SECURITY_PROTOCOL_CONFIG, "SASL_PLAINTEXT");
         * props.put(SaslConfigs.SASL_MECHANISM, "PLAIN");
         * props.put(ConsumerConfig.SESSION_TIMEOUT_MS_CONFIG, 30000);
         */

        /*
         * For SASL SCRAM mechanism instances, uncomment:
         *
         * props.put(CommonClientConfigs.SECURITY_PROTOCOL_CONFIG, "SASL_PLAINTEXT");
         * props.put(SaslConfigs.SASL_MECHANISM, "SCRAM-SHA-256");
         * props.put(ConsumerConfig.SESSION_TIMEOUT_MS_CONFIG, 30000);
         */

        // Maximum number of records per poll
        // Avoid setting too large to prevent rebalance delays
        props.put(ConsumerConfig.MAX_POLL_RECORDS_CONFIG, 30);

        // Deserialization
        props.put(ConsumerConfig.KEY_DESERIALIZER_CLASS_CONFIG,
                "org.apache.kafka.common.serialization.StringDeserializer");
        props.put(ConsumerConfig.VALUE_DESERIALIZER_CLASS_CONFIG,
                "org.apache.kafka.common.serialization.StringDeserializer");

        // Consumer group ID
        props.put(ConsumerConfig.GROUP_ID_CONFIG,
                kafkaProperties.getProperty("group.id"));

        // Create Kafka consumer instance
        KafkaConsumer<String, String> consumer =
                new KafkaConsumer<>(props);

        // Subscribe to topics (supports multiple topics)
        List<String> subscribedTopics = new ArrayList<>();

        String topicStr = kafkaProperties.getProperty("topic");
        String[] topics = topicStr.split(",");

        for (String topic : topics) {
            subscribedTopics.add(topic.trim());
        }

        consumer.subscribe(subscribedTopics);

        // Start consuming messages
        while (true) {
            try {
                ConsumerRecords<String, String> records = consumer.poll(1000);

                // Process records within session timeout
                for (ConsumerRecord<String, String> record : records) {
                    System.out.println(String.format(
                            "Consume partition:%d offset:%d",
                            record.partition(),
                            record.offset()));
```

```
                }

            } catch (Exception e) {
                try {
                    Thread.sleep(1000);
                } catch (Throwable ignore) {
                }
                e.printStackTrace();
            }
        }
    }
}
```

4.2 Golang Example

```
// kafka.json
{
  "topic": "XXX",
  "group.id": "XXX",
  "bootstrap.servers": "XXX:XX,XXX:XX,XXX:XX",
  "security.protocol": "sasl_ssl",
  "sasl.mechanism": "XXX",
  "sasl.username": "XXX",
  "sasl.password": "XXX"
}
```

```
// main.go
package main
 import (

 "encoding/json"

 "fmt"

 "github.com/confluentinc/confluent-kafka-go/kafka"

 "os"

 "path/filepath"

 )

 type KafkaConfig struct {

 Topic      string json:"topic"

 GroupId    string json:"group.id"

 BootstrapServers    string json:"bootstrap.servers"

 SecurityProtocol string json:"security.protocol"

 SaslMechanism string json:"sasl.mechanism"

 SaslUsername string json:"sasl.username"

 SaslPassword string json:"sasl.password"

 }

 // config should be a pointer to structure, if not, panic

 func loadJsonConfig() *KafkaConfig {

 workPath, err := os.Getwd()

 if err != nil {

 panic(err)

 }

 configPath := filepath.Join(workPath, "conf")

 fullPath := filepath.Join(configPath, "kafka.json")

 file, err := os.Open(fullPath);

 if (err != nil) {

 msg := fmt.Sprintf("Can not load config at %s. Error: %v", fullPath, err)

 panic(msg)

 }
```

```
defer file.Close()

decoder := json.NewDecoder(file)
var config = &amp;amp;KafkaConfig{}
err = decoder.Decode(config);
if (err != nil) {
    msg := fmt.Sprintf("Decode json fail for config file at %s. Error: %v", fullPath, err)
    panic(msg)
}
json.Marshal(config)
return  config
```

```
}


func doInitConsumer(cfg *KafkaConfig) *kafka.Consumer {

                fmt.Print("init kafka consumer, it may take a few seconds to init the connection\n")

                //common arguments

                var kafkaconf = &kafka.ConfigMap{

                "api.version.request": "true",

                "auto.offset.reset": "latest",

                "heartbeat.interval.ms": 3000,

                "session.timeout.ms": 30000,

                "max.poll.interval.ms": 120000,

                "fetch.max.bytes": 1024000,

                "max.partition.fetch.bytes": 256000}

                kafkaconf.SetKey("bootstrap.servers", cfg.BootstrapServers);

                kafkaconf.SetKey("group.id", cfg.GroupId)
```

```
switch cfg.SecurityProtocol {
case "plaintext" :
    kafkaconf.SetKey("security.protocol", "plaintext");
case "sasl_ssl":
    kafkaconf.SetKey("security.protocol", "sasl_ssl");
    kafkaconf.SetKey("ssl.ca.location", "./conf/ca-cert.pem");
    kafkaconf.SetKey("sasl.username", cfg.SaslUsername);
    kafkaconf.SetKey("sasl.password", cfg.SaslPassword);
    kafkaconf.SetKey("sasl.mechanism", cfg.SaslMechanism);
    kafkaconf.SetKey("ssl.endpoint.identification.algorithm", "None");
    kafkaconf.SetKey("enable.ssl.certificate.verification", "false")
case "sasl_plaintext":
    kafkaconf.SetKey("security.protocol", "sasl_plaintext");
    kafkaconf.SetKey("sasl.username", cfg.SaslUsername);
    kafkaconf.SetKey("sasl.password", cfg.SaslPassword);
    kafkaconf.SetKey("sasl.mechanism", cfg.SaslMechanism)

default:
    panic(kafka.NewError(kafka.ErrUnknownProtocol, "unknown protocol", true))
}

consumer, err := kafka.NewConsumer(kafkaconf)
if err != nil {
    panic(err)
}
fmt.Print("init kafka consumer success\n")
return consumer;
```

```
}


func main() {
```

```
// Choose the correct protocol
// 9092 for PLAINTEXT
// 9093 for SASL_SSL, need to provide sasl.username and sasl.password
// 9094 for SASL_PLAINTEXT, need to provide sasl.username and sasl.password
cfg := loadJsonConfig();
consumer := doInitConsumer(cfg)

consumer.SubscribeTopics([]string{cfg.Topic}, nil)

for {
    msg, err := consumer.ReadMessage(-1)
    if err == nil {
        fmt.Printf("Message on %s: %s\n", msg.TopicPartition, string(msg.Value))
    } else {
        // The client will automatically try to recover from all errors.
        fmt.Printf("Consumer error: %v (%v)\n", err, msg)
    }
}

consumer.Close()
```

```
}
```

4.3 Python Example

```
kafka_setting = {
    'sasl_plain_username': 'XXX',
    'sasl_plain_password': 'XXX',
    'bootstrap_servers': ["XXX", "XXX", "XXX"],
    'topic_name': 'XXX',
    'consumer_id': 'XXX'
}
```

```
from confluent_kafka import Consumer, KafkaError
 import setting

 conf = setting.kafka_setting

 c = Consumer({

 'bootstrap.servers': conf['bootstrap_servers'],

 'group.id': conf['group_name'],

 'auto.offset.reset': 'latest'

 })

 c.subscribe([conf['topic_name']])

 while True:

 msg = c.poll(1.0)

 if msg is None:
     continue
 if msg.error():
     if msg.error().code() == KafkaError._PARTITION_EOF:
         continue
     else:
         print("Consumer error: {}".format(msg.error()))
         continue

 print('Received message: {}'.format(msg.value().decode('utf-8')))



 c.close()
```

5. FAQ

5.1 Connection Failure

Check:

- Whether the port is open
- Whether the SASL configuration is correct

5.2 No Data Consumed

Check:

- Whether the Topic is correct
- Whether the Consumer Group conflicts
- Whether the offset is at the end (latest)

5.3 Permission Issues

Symptom:

```
Topic authorization failed
```

Solution:

```
Contact us to enable ACL permissions
```

Power Station

Create power station

Basic information

Path /op/v0/plant/create

Method POST

Interface description

request body example:

```
{
  "devices": [
    {
      "sn": "*******"
    }
  ],
  "pileSN": "*************",
  "details": {
    "name": "name_51b2289bf18f",
    "type": 1,
    "countryCode": "CN",
    "city": "city_54049a6aad08",
    "address": "address_efdf02e2819d",
    "postcode": "213414",
    "price": 0.52,
    "currency": "CNY(¥)",
    "systemCapacity": 72.00
  },
  "timezone": "Asia/Shanghai",
  "position": {
    "x": "x_9e0c3cf37a91",
    "y": "y_6fa583b5fe21",
    "format": "format_b9434db57b54",
    "pid": "pid_169c8579050c"
  },
  "electricmeterSN": "**********************",
  "layoutByMini": {
    "direction": 1,
    "arrange": [
      {
        "sn": "***********",
        "horizontal": 2,
        "vertical": 3,
        "azimuth": 30,
        "dipAngle": 45,
        "direction": 1,
        "capacity": 30,
        "model": "model_71300e86ba58"
      }
    ]
  }
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
devices	object []	Not Required		Information of modules needed for creating pv/energy-storage power station	item Type: object
├─ sn	string	Not Required		Serial number of module	
├─ key	string	Not Required			
pileSN	string	Not Required		Charge pile's serial number needed for charge pile power station	
details	object	Required		Details of power station	
├─ name	string	Required		Name of power station	
├─ type	number	Required		Type of power station 1-pv power station,2-energy-storage power station,3-charge-pile power station,4-mini-device power station	
├─ countryCode	string	Required		Country's code of power station Two-bit code, such as China is "CN",Reference address https://en.wikipedia.org/wiki/ISO_3166-1_alpha-2)	
├─ city	string	Required		City	
├─ address	string	Required		Address	
├─ postcode	string	Required		Postcode	
├─ price	number	Not Required		Electric price of power station needed for creating pv/energy-storage/mini-device power station	
├─ currency	string	Not Required		The currency of electricity price measurement needed for editing pv/energy-storage/mini-device power station reference U.S. dollar-USD($) Euro-EUR(€) Vietnamese Dong-VND(D.) RMB-CNY(¥) Pound sterling-GBP(£) Australian dollar-AUD($) Poland-PLN(Zł) India-INR(₹) Brazil-BRL(R$) S.Africa-ZAR(R) THB (THB)-THB(฿) Pakistani rupee-PKR(Rs)	
├─ systemCapacity	number	Not Required		System's capacity of power station needed for creating pv/energy-storage/mini-device power station	
timezone	string	Required		Detailed time zone in English "district"+"/"+"country" such as Asia/Shanghai Reference address https://nodatime.org/TimeZones	
position	object	Not Required		Coordinate information	
├─ format	string	Not Required		Coordinate address	
├─ x	string	Not Required		Longitude	

Name	Type	Is it required	Default value	Note	Other information
├─ y	string	Not Required		Latitude	
├─ pid	string	Not Required		Ip of address	
electricmeterSN	string	Not Required		Serial number of electricmeter needn't if empty	
layoutByMini	object	Not Required		Information of mini-device (needed for creating mini-device power station)	
├─ direction	number	Not Required		Component layout 1-portrait 2-crosswise	
├─ arrange	object []	Not Required			item Type: object
├─ sn	string	Not Required		Serial number of module (needed for creating mini-device power station)	
├─ horizontal	number	Not Required		Horizontal number (needed for creating mini-device power station)	
├─ vertical	number	Not Required		Vertical number (needed for creating mini-device power station)	
├─ azimuth	number	Not Required		Azimuth angle (needed for creating mini-device power station between 0 and 360°)	
├─ dipAngle	number	Not Required		Angle of inclination (needed for creating mini-device power station between 0 and 90°)	
├─ direction	number	Not Required		Direction needed for creating mini-device power station 1-vertical 2-horizontal	
├─ capacity	number	Not Required		Component capacity (needed for creating mini-device power station)	
├─ model	string	Not Required		Component type (needed for creating mini-device power station)	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Not Required		Error number When the result is not equal to zero, the request fails	
result	object	Not Required			
├─ stationID	string	Not Required		Id of power station	
├─ devices	object []	Not Required		Information of module	item Type: object
├─ device	object	Not Required			
├─ sn	string	Not Required		Snserial number of module	
├─ key	string	Not Required			
├─ errno	number	Not Required		Error number of module when the result is not equal to zero, the serial number cannt meet the requirements	

Delete power station

Basic information

Path /op/v0/plant/delete

Method POST

Interface description

request body example:

```
{
  "stationID": "***********"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
stationID	string	Required		Id of power station	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	

Edit power station

Basic information

Path /op/v0/plant/update

Method POST

Interface description

request body example:

```
{
  "devices": [
    {
      "sn": "*******"
    }
  ],
  "pileSN": "*************",
  "details": {
    "stationID": "********************",
    "name": "name_51b2289bf18f",
    "type": 1,
    "countryCode": "CN",
    "city": "city_54049a6aad08",
    "address": "address_efdf02e2819d",
    "postcode": "213414",
    "price": 0.52,
    "currency": "CNY(¥)",
    "systemCapacity": 72.00
  },
  "timezone": "Asia/Shanghai",
  "position": {
    "x": "x_9e0c3cf37a91",
    "y": "y_6fa583b5fe21",
    "format": "format_b9434db57b54",
    "pid": "pid_169c8579050c"
  },
  "electricmeterSN": "**********************",
  "layoutByMini": {
    "direction": 1,
    "arrange": [
      {
        "sn": "***********",
        "horizontal": 2,
        "vertical": 3,
        "azimuth": 30,
        "dipAngle": 45,
        "direction": 1,
        "capacity": 30,
        "model": "model_71300e86ba58"
      }
    ]
  }
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
devices	object []	Not Required		Information of modules needed for editing pv/energy-storage power station	item Type: object
├─ sn	string	Not Required		Serial number of module	
├─ key	string	Not Required			
pileSN	string	Not Required		Charge pile's serial number needed for charge pile power station	
details	object	Required		Details of power station	
├─ name	string	Required		Name of power station	
├─ type	number	Required		Type of power station 1-pv power station,2-energy-storage power station,3-charge-pile power station,4-mini-device power station	
├─ countryCode	string	Required		Country's code of power station Two-bit code, such as China is "CN",Reference address https://en.wikipedia.org/wiki/ISO_3166-1_alpha-2)	
├─ city	string	Required		City	
├─ address	string	Required		Address	
├─ postcode	string	Required		Postcode	
├─ price	number	Not Required		Electric price of power station needed for editing pv/energy-storage/mini-device power station	
├─ currency	string	Not Required		The currency of electricity price measurement needed for editing pv/energy-storage/mini-device power station reference U.S. dollar-USD($) Euro-EUR(€) Vietnamese Dong-VND(D.) RMB-CNY(¥) Pound sterling-GBP(£) Australian dollar-AUD($) Poland-PLN(Zł) India-INR(₹) Brazil-BRL(R$) S.Africa-ZAR(R) THB (THB)-THB(฿) Pakistani rupee-PKR(Rs)	
├─ stationID	string	Required		Id of power station	
├─ systemCapacity	number	Not Required		System's capacity of power station needed for editing pv/energy-storage/mini-device power station	
timezone	string	Required		Detailed time zone in English "district"+"/"+"country" such as Asia/Shanghai Reference address https://nodatime.org/TimeZones	
position	object	Not Required		Coordinate information	
├─ format	string	Not Required		Coordinate address	

Name	Type	Is it required	Default value	Note	Other information
├─ x	string	Not Required		Longitude	
├─ y	string	Not Required		Latitude	
├─ pid	string	Not Required		Ip of address	
electricmeterSN	string	Not Required		Serial number of electricmeter needn't if empty	
layoutByMini	object	Not Required		Information of mini-device (needed for editing mini-device power station)	
├─ direction	number	Not Required		Component layout 1-portrait 2-crosswise	
├─ arrange	object []	Not Required			item Type: object
├─ sn	string	Not Required		Serial number of module	
├─ horizontal	number	Not Required		Horizontal number	
├─ vertical	number	Not Required		Vertical number	
├─ azimuth	number	Not Required		Azimuth angle(between 0 and 360°	
├─ dipAngle	number	Not Required		Angle of inclination between 0 and 90°	
├─ direction	number	Not Required		Direction 1-vertical 2-horizontal	
├─ capacity	number	Not Required		Component capacity	
├─ model	string	Not Required		Component type	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Not Required		Error number When the result is not equal to zero, the request fails	
result	object	Not Required			
├─ stationID	string	Not Required		Id of power station	
├─ devices	object []	Not Required		Information of module	item Type: object
├─ device	object	Not Required			
├─ sn	string	Not Required		Serial number of module	
├─ key	string	Not Required			
├─ errno	number	Not Required		Error number of module when the result is not equal to zero, the serial number cannt meet the requirements	

Get power station detail

Basic information

Path /op/v0/plant/detail

Method GET

Interface description

Obtain details about a power station based on the power station ID

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
id	Required		Id of power station

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ stationName	string	Required		Name of power station	
├─ country	string	Required		Country of power station	
├─ city	string	Required		City of power station	
├─ address	string	Required		Address of power station	
├─ createDate	string	Required		Create time,millisecond timestamp	
├─ postcode	string	Required		Postcode of power station	
├─ capacity	number	Required		Installed capacity (Unit kw)	
├─ timezone	string	Required		Time zone of power station	
├─ user	object	Required		User information	
├─ name	string	Required		Name of user	
├─ email	string	Required		Email of user	
├─ phone	string	Required		Phone of user	
├─ installer	object	Required		Installer information	
├─ name	string	Required		Name of Installer	
├─ email	string	Required		Email of Installer	
├─ phone	string	Required		Phone of Installer	
├─ modules	object []	Required		Device list	item Type: object
├─ moduleSN	string	Required		Serial number of Data Logger	
├─ deviceSN	string	Required		Serial number of Inverter	
├─ batteries	object []	Required		Battery list (US Only)	item Type: object
├─ batterySN	string	Required		Serial number of the battery	
├─ type	string	Required		Master Slave Type of the battery	
├─ version	string	Required		Version of the battery	
├─ model	string	Required		model of the battery	
├─ capacity	integer	Required		capicty of the battery	
├─ productDate	integer	Required		Manufacture date of the battery The date of manufacture (time portion ignored), expressed as a Unix timestamp in milliseconds	

Get power station list

Basic information

Path /op/v0/plant/list

Method POST

Interface description

Obtain the list of power stations to which this account belongs

request body example:

```
{
  "currentPage": 1,
  "pageSize": 10
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ data	object []	Required			item Type: object
├─ stationID	string	Required		Id of power station	
├─ name	string	Required		Name of power station	
├─ ianaTimezone	string	Required		Timezone of power station	
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	

VPP

authorization code mode offboard by sn

Basic information

Path /op/v0/vpp/oauth2/code/offboard/device

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	

client mode offboard

Basic information

Path /op/v0/vpp/oauth2/client/offboard

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	

client mode onboard

Basic information

Path /op/v0/vpp/oauth2/client/onboard

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	

Inverter-V1

Get device detail

Basic information

Path /op/v1/device/detail

Method GET

Interface description

Obtain inverter details

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type		Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Serial number of Inverter

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ deviceSN	string	Required		Serial number of the inverter	
├─ moduleSN	string	Required		Serial number of the collector	
├─ stationID	string	Required		Id of power station	
├─ stationName	string	Required		Name of power station	
├─ afciVersion	string	Required		Version of afci	
├─ managerVersion	string	Required		Version of manager	
├─ masterVersion	string	Required		Version of master	
├─ slaveVersion	string	Required		Version of slave	
├─ hardwareVersion	string	Required		Version of hardware	
├─ hubVersion	string	Required		Version of hub (US)	
├─ status	integer	Required		Status of device Enumeration : 1:online 2:breakdown 3:offline	Enumeration: 1,2,3 Enumeration note: 1:online 2:breakdown 3:offline
├─ capacity	number	Required		Capacity of device(in kW)	
├─ thirdPartyGen	boolean	Required		Does the equipment connect to the third-party power generation equipment	
├─ function	object	Required		Functional support	
├─ scheduler	boolean	Required		Whether the periodic mode switchover function is supported	
├─ batteryList	object[]	Required			item Type: object
├─ batterySN	string	Required		Serial number of the battery	
├─ type	string	Required		Master Slave Type of the battery	
├─ version	string	Required		Version of the battery	
├─ model	string	Required		model of the battery	
├─ capicty	integer	Required		capicty of the battery	
├─ productDate	integer	Required		Manufacture date of the battery The date of manufacture (time portion ignored), expressed as a Unix timestamp in milliseconds	

Get device real-time data

Basic information

Path /op/v1/device/real/query

Method POST

Interface description

Get the latest real-time data of the inverter. The sn array is mandatory and returns only the result set belonging to the current user. If no variable is specified, all variables are queried

request body example:

```
{
  "sns": ["***************","***************","***************"],
  "variables": ["pvPower","pv1Current"]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
variables	string []	Not Required		If this parameter is not passed, all variable data is obtained by default item type: string	item Type: string
├─		Not Required		Variable name Available variables can be queried through the Get Variables interface Remark:Variable name,available variables can be queried through the Get Variables interface	
sns	string []	Required		Serial Number of Inverter Remark:Can transmit up to 50 Serial Number	item Type: string
├─		Not Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object []	Required			item Type: object
├─ deviceSN	string	Required		Serial Number of Inverter	
├─ datas	object []	Required			item Type: object
├─ variable	string	Required		Variable name If the data is not found, it will not be returned	
├─ unit	string	Required		Unit	
├─ name	string	Required		Name in English	
├─ value	number	Required		Value	
├─ time	string	Required		Inverter local time, yyyy-MM-dd HH:mm:ss zZ format	

Inverter

Get device list

Basic information

Path /op/v0/device/list

Method POST

Interface description

Obtain the list of inverters owned by this account

request body example:

```
{
  "currentPage": 1,
  "pageSize": 10
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ deviceSN	string	Required		Serial number of the inverter	
├─ moduleSN	string	Required		Serial number of the Data Logger	
├─ stationID	string	Required		Id of power station	
├─ stationName	string	Required		Name of power station	
├─ status	integer	Required		Status of module	Enumeration: 1,2,3 Enumeration note: 1:online 2:fault 3:offline
├─ hasPV	boolean	Required		Is there a photovoltaic system available	
├─ hasBattery	boolean	Required		Is there a battery available	
├─ deviceType	string	Required		Model of device	
├─ productType	string	Required		Series of product	

Get device detail (Deprecated)

Basic information

Path /op/v0/device/detail

Method GET

Interface description

Obtain inverter details

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Serial number of Inverter

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ deviceSN	string	Required		Serial number of the inverter	
├─ moduleSN	string	Required		Serial number of the collector	
├─ stationID	string	Required		Id of power station	
├─ stationName	string	Required		Name of power station	
├─ afciVersion	string	Required		Version of afci	
├─ managerVersion	string	Required		Version of manager	
├─ masterVersion	string	Required		Version of master	
├─ slaveVersion	string	Required		Version of slave	
├─ hardwareVersion	string	Required		Version of hardware	
├─ status	integer	Required		Status of device	Enumeration: 1,2,3 Enumeration note: 1:online 2:breakdown 3:offline
├─ capacity	integer	Required		Capacity of device(in kW)	
├─ function	object	Required		Functional support	
├─ scheduler	boolean	Required		Whether the periodic mode switchover function is supported	
├─ batteryList	object []	Required			item Type: object
├─ batterySN	string	Required		Serial number of the battery	
├─ type	string	Required		Master Slave Type of the battery	
├─ version	string	Required		Version of the battery	
├─ model	string	Required		model of the battery	
├─ capicty	string	Required		capicty of the battery	

Get available variables

Basic information

Path /op/v0/device/variable/get

Method GET

Interface description

Obtain variable tables for use by real-time data and historical data interfaces

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
	string	Not Required			format: binary

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object []	Required			item Type: object
├─ {variable}	object	Required			
├─ unit	string	Required		unit	
├─ name	object	Required		variable name	
├─ zh_CN	string	Required			
├─ en	string	Required			
├─ de	string	Required			
├─ pt	string	Required			
├─ fr	string	Required			
├─ pl	string	Required			

Get error code information

Basic information

Path /op/v0/device/fault/get

Method GET

Interface description

Obtain error code information

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
	string	Not Required			format: binary

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object []	Required			item Type: object
├─ {errNo}	object	Required		Error code	
├─ en	string	Required		The error code corresponds in English	
├─ zh_CN	string	Required		The error code corresponds in Chinese	
├─ pl	string	Required		The error code corresponds in Polish	

Get device real-time data (Deprecated)

Basic information

Path /op/v0/device/real/query

Method POST

Interface description

Get the latest real-time data of the inverter, and get the real-time data of all devices under the name without specifying the SN

request body example:

```
{
  "sn": "***************",
  "variables": ["pvPower","pv1Current"]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
variables	string []	Not Required		If this parameter is not passed, all variable data is obtained by default item type: string	item Type: string
├─		Not Required		Variable name Available variables can be queried through the Get Variables interface Remark:Variable name,available variables can be queried through the Get Variables interface	
sn	string	Not Required		Serial Number of Inverter If no parameter is specified, all devices are obtained	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object []	Required			item Type: object
├─ deviceSN	string	Required		Serial Number of Inverter	
├─ datas	object []	Required			item Type: object
├─ variable	string	Required		Variable name If the data is not found, it will not be returned	
├─ unit	string	Required		Unit	
├─ name	string	Required		Name in English	
├─ value	number	Required		Value	
├─ time	string	Required		Time of Data Update, utc time	

Get device history data

Basic information

Path /op/v0/device/history/query

Method POST

Interface description

The historical data within the specified 24 hours can be obtained The begin time and end time are required). If no time period is specified, you can obtain the historical data of the last three days.

request body example:

```
{
  "sn": "***************",
  "variables": ["pvPower","pv1Current"],
  "begin": 1703548800000,
  "end": 1703635200000
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
variables	string []	Not Required		If this parameter is not passed, all variable data is obtained by default	item Type: string
├─		Not Required		Variable name Available variables can be queried through the Get Variables interface Remark:Variable name,available variables can be queried through the Get Variables interface	
sn	string	Required		Serial Number of Inverter	
begin	number	Not Required		Start timestamp(milliseconds)	
end	number	Not Required		End timestamp(milliseconds)	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object []	Required			item Type: object
├─ deviceSN	string	Required		Serial Number of Inverter	
├─ datas	object []	Required			item Type: object
├─ variable	string	Required		Variable name	
├─ unit	string	Required		Unit of variable	
├─ name	string	Required		Name in English	
├─ data	object []	Required			item Type: object
├─ value	number	Required		Value	
├─ time	string	Required		Time of Data Update, utc time	

Get device production report

Basic information

Path /op/v0/device/report/query

Method POST

Interface description

Obtain the inverter electricity consumption report. Calculated according to the time zone of the power station to which the inverter belongs.The monthly electricity consumption of the year is obtained, the daily electricity consumption of the month is obtained, and the hourly electricity consumption of the day is obtained

request body example:

```
{
  "sn": "***************",
  "year": 2023,
  "month": 10,
  "day": 3,
  "dimension": "day",
  "variables": ["generation","feedin","gridConsumption","chargeEnergyToTal","dischargeEnergyToTal","PVEnergyTotal"]
}
{
  "sn": "***************",
  "year": 2023,
  "month": 10,
  "dimension": "month",
  "variables": ["generation","feedin","gridConsumption","chargeEnergyToTal","dischargeEnergyToTal","PVEnergyTotal"]
}
{
  "sn": "***************",
  "year": 2023,
  "dimension": "year",
  "variables": ["generation","feedin","gridConsumption","chargeEnergyToTal","dischargeEnergyToTal"]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of the inverter	
year	integer	Required		year	Minimum value: 2020
month	integer	Not Required		month	Maximum value: 12 Minimum value: 1
day	integer	Not Required		day	Maximum value: 31 Minimum value: 1
dimension	string	Required		Statistical dimension	Enumeration: year,month,day
variables	string []	Required		Query variable	item Type: string
├─		Not Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object []	Required			item Type: object
├─ variable	string	Required		Variable name	
├─ unit	string	Required		Variable unit	
├─ values	number []	Required		Variable value	item Type: number
├─		Not Required			

Get device power generation

Basic information

Path /op/v0/device/generation

Method GET

Interface description

Obtain the energy yield information of the inverter based on the time zone of the power station to which the inverter belongs

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Serial number of inverter

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├— today	number	Required		Electricity generation of this today unit:kWh	
├— month	number	Required		Electricity generation of this month unit:kWh	
├— cumulative	number	Required		Cumulative power generation unit:kWh	

Get the minimum soc settings for the battery of device

Basic information

Path /op/v0/device/battery/soc/get

Method GET

Interface description

Obtain the inverter battery soc Settings

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Serial number of Inverter

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ minSoc	integer	Required		Minimum soc of system	
├─ minSocOnGrid	integer	Required		Battery discharge cutoff SOC in grid-connected state	

Set the minmum soc for the battery of the device

Basic information

Path /op/v0/device/battery/soc/set

Method POST

Interface description

Set the minmum soc for the battery of the device

request body example:

```
{
  "sn": "***************",
  "minSoc": 10,
  "minSocOnGrid": 10
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of inverter	
minSoc	integer	Required		Minimum soc of system	Maximum value: 100 Minimum value: 10
minSocOnGrid	integer	Required		Battery discharge cutoff SOC in grid-connected state	Maximum value: 100 Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
result	object	Required			

Get the setting of battery charging time

Basic information

Path /op/v0/device/battery/forceChargeTime/get

Method GET

Interface description

Obtain the setting of battery charging time

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Serial number of inverter

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ enable1	string	Required		Whether the first segment is enabled	
├─ startTime1	object	Required		The first period starts time	
├─ hour	integer	Required			
├─ minute	integer	Required			
├─ endTime1	object	Required		The end time of the first period	
├─ hour	integer	Required			
├─ minute	integer	Required			
├─ enable2	string	Required		Whether the second segment is enabled or not	
├─ startTime2	object	Required		The second period starts time	
├─ hour	integer	Required			
├─ minute	integer	Required			
├─ endTime2	object	Required		The end time of the second period	
├─ hour	integer	Required			
├─ minute	integer	Required			

Set the battery charging time

Basic information

Path /op/v0/device/battery/forceChargeTime/set

Method POST

Interface description

Set the battery charging time. The period must be two periods. Both periods must be delivered

request body :

```
{
  "sn": "sn_4e50c9b8a5c3",
  "enable1": false,
  "enable2": false,
  "startTime1": {
    "hour": 0,
    "minute": 0
  },
  "endTime1": {
    "hour": 0,
    "minute": 0
  },
  "startTime2": {
    "hour": 0,
    "minute": 0
  },
  "endTime2": {
    "hour": 0,
    "minute": 0
  }
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
enable1	boolean	Required		Whether the first segment is enabled	
enable2	boolean	Required		Whether the second segment is enabled or not	
startTime1	object	Required		The first period starts time	
├— hour	integer	Required			Maximum value: 23 Minimum value: 0
├— minute	integer	Required			Maximum value: 59 Minimum value: 0
endTime1	object	Required		The end time of the first period	
├— hour	integer	Required			Maximum value: 23 Minimum value: 0
├— minute	integer	Required			Maximum value: 59 Minimum value: 0
startTime2	object	Required		The second period starts time	
├— hour	integer	Required			Maximum value: 23 Minimum value: 0
├— minute	integer	Required			Maximum value: 59 Minimum value: 0
endTime2	object	Required		The end time of the second period	
├— hour	integer	Required			Maximum value: 23 Minimum value: 0
├— minute	integer	Required			Maximum value: 59 Minimum value: 0

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Serial number of Inverter	
├─ enable1	boolean	Not Required		Whether the first segment is enabled	
├─ enable2	boolean	Not Required		Whether the second segment is enabled or not	
├─ startTime1	object	Not Required		The first period starts time	
├─ hour	integer	Required			Maximum value: 23 Minimum value: 0
├─ minute	integer	Required			Maximum value: 59 Minimum value: 0
├─ endTime1	object	Not Required		The end time of the first period	
├─ hour	integer	Required			Maximum value: 23 Minimum value: 0
├─ minute	integer	Required			Maximum value: 59 Minimum value: 0
├─ startTime2	object	Not Required		The second period starts time	
├─ hour	integer	Required			Maximum value: 23 Minimum value: 0
├─ minute	integer	Required			Maximum value: 59 Minimum value: 0
├─ endTime2	object	Not Required		The end time of the second period	
├─ hour	integer	Required			Maximum value: 23 Minimum value: 0
├─ minute	integer	Required			Maximum value: 59 Minimum value: 0

Get the device settings item

Basic information

Path /op/v0/device/setting/get

Method POST

Interface description

Obtain the device settings

request body :

```
{
  "sn": "sn_4e50c9b8a5c3",
  "key": "ExportLimit"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
key	string	Required		settings item	Enumeration: ExportLimit,MinSoc,MinSocOnGrid,MaxSoc,GridCode,WorkMode,ActivePowerLimit,ExportLimitPower,EpsOutPut,MaxSetChargeCurrent,MaxSetDischargeCurrent,ECOMode,Meter1Enable,Meter2Enable,Sy

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ value	string	Required		setting value	
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	

Set the device settings item

Basic information

Path /op/v0/device/setting/set

Method POST

Interface description

Deliver device settings

request body :

```
{
  "sn": "sn_4e50c9b8a5c3",
  "key": "ExportLimit",
  "value": "13000"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
key	string	Required		settings item	Enumeration: ExportLimit,MinSoc,MinSocOnGrid,MaxSoc,GridCode,WorkMode,ActivePowerLimit,ExportLimitPower,EpsOutPut,MaxSetChargeCurrent,MaxSetDischargeCurrent,ECOMode,Meter1Enable,Meter2Enable,Sy
value	string	Required		settings value	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Serial number of Inverter	
├─ key	string	Not Required		settings item	Enumeration: ExportLimit,MinSoc,MinSocOnGrid,MaxSoc,GridCode,WorkMode,ActivePowerLimit,ExportLimitPower,EpsOutPut,MaxSetChargeCurrent,MaxSetDischargeCurrent,ECOMode,Meter1Enable,Meter2Enable
├─ value	string	Not Required		settings value	

Get the device time

Basic information

Path /op/v0/device/time/get

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object	Required			
├─ year	string	Required			
├─ month	string	Required			
├─ day	string	Required			
├─ hour	string	Required			
├─ second	string	Required			
├─ minute	string	Required			

Set the device time

Basic information

Path /op/v0/device/time/set

Method POST

Interface description

Modify the device time

Note that the device's automatic time synchronization function will be disabled after modification

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
year	string	Required			
second	string	Required		0-59	
minute	string	Required		0-59	
hour	string	Required		0-23	
day	string	Required		1-31	
month	string	Required		1-12	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Serial number of Inverter	
├─ year	string	Not Required			
├─ second	string	Not Required		0-59	
├─ minute	string	Not Required		0-59	
├─ hour	string	Not Required		0-23	
├─ day	string	Not Required		1-31	
├─ month	string	Not Required		1-12	

Get the device peakshaving settings

Basic information

Path /op/v0/device/peakShaving/get

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ importLimit	object	Required			
├─ value	string	Required		setting value	
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	
├─ soc	object	Required			
├─ value	string	Required		setting value	
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	

Set the device peakshaving settings

Basic information

Path /op/v0/device/peakShaving/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
importLimit	number	Required			
soc	integer	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Serial number of Inverter	
├─ importLimit	number	Not Required			
├─ soc	integer	Not Required			

Get the Meter Reader

Basic information

Path /op/v0/device/getMeterReader

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ readerType	number	Required		Electric meter reader type	Enumeration: 0,1,2,3,4,5,6,8,9,10,11,12 Enumeration note: Close meter reader connection, SolarmanPV, EARN-E, SHELLY, WIZARD, Chint, AECC, iometer, Shelly Pro EM-50, Powerfox, Shelly 3EM, Solakon
├─ readerStatus	number	Required		Electric meter reader status	Enumeration: 0,1 Enumeration note: Electric meter data not obtained, Electric meter data has been obtained
├─ readerInfo	object	Required		Electric meter reader info	
├─ ip	string	Not Required		ip address	
├─ key	string	Not Required		password for meter	

Set the Meter Reader

Basic information

Path /op/v0/device/setMeterReader

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number	
readerType	number	Required		Electric meter reader type	Enumeration: 0,1,2,3,4,5,6,8,9,10,11,12 Enumeration note: Close meter reader connection, SolarmanPV, EARN-E, SHELLY, WIZARD, Chint, AECC, iometer, Shelly Pro EM-50, Powerfox, Shelly 3EM, Solakon
readerInfo	object	Required		Electric meter reader info	
├─ ip	string	Not Required		ip address	
├─ key	string	Not Required		password for meter	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Serial number	
├─ readerType	number	Not Required		Electric meter reader type	Enumeration: 0,1,2,3,4,5,6,8,9,10,11,12 Enumeration note: Close meter reader connection, SolarmanPV, EARN-E, SHELLY, WIZARD, Chint, AECC, iometer, Shelly Pro EM-50, Powerfox, Shelly 3EM, Solakon
├─ readerInfo	object	Not Required		Electric meter reader info	
├─ ip	string	Not Required		ip address	
├─ key	string	Not Required		password for meter	

Get battery heating Parameter

Basic information

Path /op/v0/device/batteryHeating/get

Method POST

Interface description

response body example:


```
{
  "errno": 0,
  "msg": "Operation successful",
  "result": {
    "dataList": [
      {
        "unit": "",
        "precision": 1.0,
        "name": "time1EndHour",
        "range": {
          "min": 0.0,
          "max": 23.0
```

```
    },
    "value": "0"
},
{
    "unit": "",
    "precision": 1.0,
    "name": "minEndTemperatureRange",
    "value": "10"
},
{
    "unit": "",
    "precision": 1.0,
    "name": "maxStartTemperatureRange",
    "value": "9"
},
{
    "unit": "",
    "precision": 1.0,
    "name": "time2StartHour",
    "range": {
        "min": 0.0,
        "max": 23.0
    },
    "value": "0"
},
{
    "unit": "",
    "precision": 1.0,
    "name": "time1EndMinute",
    "range": {
        "min": 0.0,
        "max": 59.0
    },
    "value": "0"
},
{
    "unit": "",
    "precision": 1.0,
    "name": "time3StartHour",
    "range": {
        "min": 0.0,
        "max": 23.0
    },
```

```
    "value": "0"
  },
  {
    "unit": "",
    "precision": 1.0,
    "name": "minStartTemperatureRange",
    "value": "1"
  },
  {
    "enumList": [
      "disable",
      "enable"
    ],
    "unit": "°C",
    "precision": 1.0,
    "name": "time2Enable",
    "value": "disable"
  },
  {
    "enumList": [
      "disable",
      "enable"
    ],
    "unit": "",
    "precision": 1.0,
    "name": "batteryWarmUpFunctionEnableFlag",
    "value": "disable"
  },
  {
    "unit": "",
    "precision": 1.0,
    "name": "time3EndMinute",
    "range": {
      "min": 0.0,
      "max": 59.0
    },
    "value": "0"
  },
  {
    "unit": "",
    "precision": 1.0,
    "name": "time2EndHour",
    "range": {
```

```
        "min": 0.0,
        "max": 23.0
      },
      "value": "0"
    },
    {
      "enumList": [
        "disable",
        "enable"
      ],
      "unit": "°C",
      "precision": 1.0,
      "name": "time3Enable",
      "value": "disable"
    },
    {
      "unit": "",
      "precision": 1.0,
      "name": "time1StartHour",
      "range": {
        "min": 0.0,
        "max": 23.0
      },
      "value": "0"
    },
    {
      "unit": "°C",
      "precision": 1.0,
      "name": "startTemperature",
      "value": "1"
    },
    {
      "enumList": [
        "disable",
        "enable"
      ],
      "unit": "",
      "precision": 1.0,
      "name": "time1Enable",
      "value": "disable"
    },
    {
      "enumList": [
```

```
        "disable",
        "enable"
      ],
      "unit": "",
      "precision": 1.0,
      "name": "batteryWarmUpEnable",
      "value": "disable"
    },
    {
      "unit": "°C",
      "precision": 1.0,
      "name": "endTemperature",
      "value": "10"
    },
    {
      "unit": "",
      "precision": 1.0,
      "name": "time3EndHour",
      "range": {
        "min": 0.0,
        "max": 23.0
      },
      "value": "0"
    },
    {
      "unit": "",
      "precision": 1.0,
      "name": "maxEndTemperatureRange",
      "value": "15"
    },
    {
      "unit": "",
      "precision": 1.0,
      "name": "time2StartMinute",
      "range": {
        "min": 0.0,
        "max": 59.0
      },
      "value": "0"
    },
    {
      "unit": "",
      "precision": 1.0,
```

```
            "name": "time2EndMinute",
            "range": {
              "min": 0.0,
              "max": 59.0
            },
            "value": "0"
          },
          {
            "unit": "",
            "precision": 1.0,
            "name": "time1StartMinute",
            "range": {
              "min": 0.0,
              "max": 59.0
            },
            "value": "0"
          },
          {
            "enumList": [
              "The battery is in a stopped warm up state",
              "The battery is in a ready warm up state",
              "The battery is in the warm up state",
              "Battery is in self warm up state",
              "The battery is in a ready stopped warm up state"
            ],
            "unit": "",
            "precision": 1.0,
            "name": "batteryWarmUpState",
            "value": "The battery is in a stopped warm up state"
          },
          {
            "unit": "",
            "precision": 1.0,
            "name": "time3StartMinute",
            "range": {
              "min": 0.0,
              "max": 59.0
            },
            "value": "0"
          }
        ]
      }
    }
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Not Required			
msg	string	Not Required			
result	object	Not Required			
├─ dataList	object []	Not Required			item Type: object
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ name	string	Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			
├─ enumList	string []	Not Required			item Type: string
├─		Not Required			

Set battery heating Parameter

Basic information

Path /op/v0/device/batteryHeating/set

Method POST

Interface description

response body example:


```
{
  "errno": 0,
  "msg": "Operation successful",
  "result": null
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		device sn	
batteryWarmUpEnable	string	Required		Battery warm up enable	
startTemperature	string	Required		Start temperature	
endTemperature	string	Required		End temperature	
time1Enable	string	Required		Time1 enable	
time1StartHour	string	Required		Start time1 - hour	
time1StartMinute	string	Required		Start time1 - minute	
time1EndHour	string	Required		Ending time1 - hour	
time1EndMinute	string	Required		Ending time1 - minute	
time2Enable	string	Required		Time2 Enable	
time2StartHour	string	Required		Start time2 - hour	
time2StartMinute	string	Required		Start time2 - minute	
time2EndHour	string	Required		Ending time2 - hour	
time2EndMinute	string	Required		Ending time2 - minute	
time3Enable	string	Required		Time3 enable	
time3StartHour	string	Required		Start time3 - hour	
time3StartMinute	string	Required		Start time3 - minute	
time3EndHour	string	Required		Ending time3 - hour	
time3EndMinute	string	Required		Ending time3 - minute	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		device sn	
├─ batteryWarmUpEnable	string	Not Required		Battery warm up enable	
├─ startTemperature	string	Not Required		Start temperature	
├─ endTemperature	string	Not Required		End temperature	
├─ time1Enable	string	Not Required		Time1 enable	
├─ time1StartHour	string	Not Required		Start time1 - hour	
├─ time1StartMinute	string	Not Required		Start time1 - minute	
├─ time1EndHour	string	Not Required		Ending time1 - hour	
├─ time1EndMinute	string	Not Required		Ending time1 - minute	
├─ time2Enable	string	Not Required		Time2 Enable	
├─ time2StartHour	string	Not Required		Start time2 - hour	
├─ time2StartMinute	string	Not Required		Start time2 - minute	
├─ time2EndHour	string	Not Required		Ending time2 - hour	
├─ time2EndMinute	string	Not Required		Ending time2 - minute	
├─ time3Enable	string	Not Required		Time3 enable	
├─ time3StartHour	string	Not Required		Start time3 - hour	
├─ time3StartMinute	string	Not Required		Start time3 - minute	
├─ time3EndHour	string	Not Required		Ending time3 - hour	
├─ time3EndMinute	string	Not Required		Ending time3 - minute	

Get display sleep Paeameter

Basic information

Path /op/v0/device/displaySleep/get

Method POST

Interface description

response body example:

{

```
"errno": 0,
"msg": "Operation successful",
"result": {
  "displaySleepMode": {
    "unit": "",
    "precision": 1.0,
    "range": {
      "min": 0.0,
      "max": 1.0
    },
    "value": "true"
  },
  "displaySleepModeStartTimeHour": {
    "unit": "",
    "precision": 1.0,
    "range": {
      "min": 0.0,
      "max": 23.0
    },
    "value": "1"
  },
  "displaySleepModeEndTimeHour": {
    "unit": "",
    "precision": 1.0,
    "range": {
      "min": 0.0,
      "max": 23.0
    },
    "value": "5"
  },
  "displaySleepModeEndTimeMin": {
    "unit": "",
    "precision": 1.0,
    "range": {
      "min": 0.0,
      "max": 59.0
    },
    "value": "30"
  },
  "displaySleepModeStartTimeMin": {
    "unit": "",
    "precision": 1.0,
    "range": {
```

```
            "min": 0.0,
            "max": 59.0
          },
          "value": "10"
        }
      }
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		device sn	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Not Required			
msg	string	Not Required			
result	object	Not Required			
├─ displaySleepMode	object	Not Required			
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			
├─ displaySleepModeStartTimeHour	object	Not Required			
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			
├─ displaySleepModeEndTimeHour	object	Not Required			
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			
├─ displaySleepModeEndTimeMin	object	Not Required			
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			
├─ displaySleepModeStartTimeMin	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ unit	string	Not Required			
├─ precision	number	Not Required			
├─ range	object	Not Required			
├─ min	number	Not Required			
├─ max	number	Not Required			
├─ value	string	Not Required			

Set display sleep Paeameter

Basic information

Path /op/v0/device/displaySleep/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Device sn	
displaySleepMode	string	Required		Sleep mode	
displaySleepModeStartTimeHour	string	Required		The start time of SleepMode is hours	
displaySleepModeStartTimeMin	string	Required		The start time of SleepMode is minutes	
displaySleepModeEndTimeHour	string	Required		The end time of SleepMode is hours	
displaySleepModeEndTimeMin	string	Required		The end time of SleepMode is minutes	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Device sn	
├─ displaySleepMode	string	Not Required		Sleep mode	
├─ displaySleepModeStartTimeHour	string	Not Required		The start time of SleepMode is hours	
├─ displaySleepModeStartTimeMin	string	Not Required		The start time of SleepMode is minutes	
├─ displaySleepModeEndTimeHour	string	Not Required		The end time of SleepMode is hours	
├─ displaySleepModeEndTimeMin	string	Not Required		The end time of SleepMode is minutes	

Get device boarding status

Basic information

Path /op/v0/device/boarding/status

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
sn	Required		Device sn

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
msg	string	Required			
result	object	Required			
├— status	integer	Required		Represents the binding state. 0 for Pending Confirmation, 1 for Confirmed.	Enumeration: 0,1

Set device boarding status

Basic information

Path /op/v0/device/boarding/status/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Device sn	
status	number	Required		Represents the binding state. 0 for Pending Confirmation, 1 for Confirmed.	Enumeration: 0,1

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
msg	string	Required			
result	object	Required			

Get Device Fault History

Basic information

Path /op/v0/device/fault/history

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	
date	string	Required		Target date for the query (Format: YYYY-MM-DD)	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required			
result	object	Required			
├─ sn	string	Required		Serial number of Inverter	
├─ faults	object []	Required			item Type: object
├─ faultCode	string	Required		fault code	
├─ faultMessage	string	Required		fault message	
├─ timestamp	number	Required		Unix timestamp (milliseconds) when the fault occurred.	

Scheduler-V3

Batch set the time segment information_copy

Basic information

Path /op/v3/device/scheduler/enable/batch

Method POST

Interface description

This API is used to batch configure Scheduler V3 time segment settings for multiple devices asynchronously.
The system first validates all request parameters, device serial numbers, and access permissions. The validation is all-or-nothing: if any device is invalid, unauthorized, or any scheduler time segment is invalid, the entire request is rejected and no task is submitted.
After validation succeeds, the system accepts the batch request, splits it into individual device tasks, and submits them for asynchronous execution. The API returns immediately with ACCEPTED; it does not wait for all devices to finish.
If notifyChannel is none, no completion notification is sent.
If notifyChannel is webhook, a webhook notification is sent after all device tasks are completed.
Webhook Signature Verification Steps

1. Read X-Foxess-Timestamp, X-Foxess-Nonce, and X-Foxess-Signature.
2. Reject the request if any of these headers are missing.
3. Optionally reject the request if the timestamp is too old, for example older than 1 Hour.
4. Build the string to sign: `timestamp + "\n" + nonce + "\n" + rawBody`
5. Calculate HMAC-SHA256 using the webhook secret.
6. Hex-encode the result in lowercase.
7. Compare it with the value after sha256= in X-Foxess-Signature.
8. Use constant-time comparison to avoid timing attacks.
9. Use X-Foxess-Event-Id for idempotency. If the same event ID is received again, it should not be processed twice.

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
requestId	string	Not Required		Client-generated idempotency ID for this batch request. Maximum length: 64 characters. If omitted, the system generates one. Reusing the same requestId for the same client/user is treated as a duplicate request.	
notifyChannel	string	Required		Completion notification channel. Supported values: none, webhook	
deviceSNList	string []	Required		Device serial number list. The list must not be empty. All devices must exist and belong to the current user/client. If any device is invalid or unauthorized, the whole request is rejected. Current maximum: 50 devices.	item Type: string
├—		Not Required			
groups	object []	Required			item Type: object
├— startHour	integer	Required		Start time - hour	
├— startMinute	integer	Required		Start time - minute	
├— endHour	integer	Required		Ending time - hour	
├— endMinute	integer	Required		Ending time - minute	
├— workMode	string	Required		Working mode	
├— extraParam	object	Not Required			
├— minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├— fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├— fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├— maxSoc	number	Not Required		Max soc value	
├— importLimit	number	Not Required		Import Limit	
├— exportLimit	number	Not Required		Export Limit	
├— pvLimit	number	Not Required		PV Limit	
├— reactivePower	number	Not Required		Reactive Power Limit	

Name	Type	Is it required	Default value	Note	Other information
isDefault	boolean	Not Required		Whether to reset unspecified extra parameters to system defaults. false: unspecified extraParam fields remain unchanged. true: unspecified extraParam fields are restored to system defaults.	
webhook	object	Not Required		Temporary webhook configuration is used for request-level webhook overrides, primarily intended for development and debugging purposes. For enhanced security and reliability in production, we recommend registering your webhook configurations directly within our platform.	
├─ url	string	Not Required		Temporary webhook callback URL. Must be HTTPS. Required if notifyChannel=webhook	
├─ secret	string	Not Required		Temporary webhook signing secret. Required when webhook.url is used. This secret is used to generate the webhook signature.	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ requestId	string	Not Required		The idempotency ID of this batch request. If the client did not provide one, the system returns the generated ID.	
├─ status	boolean	Not Required	false	Batch request acceptance status. Current value: ACCEPTED, meaning the request has passed validation and has been submitted for asynchronous execution.	
├─ total	string	Required		Total number of devices accepted in this batch request.	
├─ accepted	string	Required		Number of device tasks successfully submitted for asynchronous processing.	

Get the time segment information

Basic information

Path /op/v3/device/scheduler/get

Method POST

Interface description

Get timing mode segment information

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ enable	string	Required		State of master switch (0:off 1:on)	
├─ groups	object []	Required			item Type: object
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ extraParam	object	Not Required			
├─ minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Not Required		Max soc value	
├─ importLimit	number	Not Required		Import Limit	
├─ exportLimit	number	Not Required		Export Limit	

Name	Type	Is it required	Default value	Note	Other information
├─ pvLimit	number	Not Required		PV Limit	
├─ reactivePower	number	Not Required		Reactive Power Limit	
├─ properties	object	Required		Range for each field in the group time period	
├─ {field_name}	object	Required			
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	
├─ maxGroupCount	number	Required		Maximum number of groups supported by the current device	

Set the time segment information

Basic information

Path /op/v3/device/scheduler/enable

Method POST

Interface description

Set the time segment information of the timing mode

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
isDefault	boolean	Not Required	false	- false: Parameters not provided in `extraParam` remain unchanged. - true: Parameters not provided in `extraParam` are restored to system defaults.	
groups	object []	Required			item Type: object
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ extraParam	object	Not Required			
├─ minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Not Required		Max soc value	
├─ importLimit	number	Not Required		Import Limit	
├─ exportLimit	number	Not Required		Export Limit	
├─ pvLimit	number	Not Required		PV Limit	

Name	Type	Is it required	Default value	Note	Other information
⊢— reactivePower	number	Not Required		Reactive Power Limit	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ isDefault	boolean	Not Required	false	- false: Parameters not provided in `extraParam` remain unchanged. - true: Parameters not provided in `extraParam` are restored to system defaults.	
├─ groups	object []	Not Required			item Type: object
├─ startHour	integer	Not Required		Start time - hour	
├─ startMinute	integer	Not Required		Start time - minute	
├─ endHour	integer	Not Required		Ending time - hour	
├─ endMinute	integer	Not Required		Ending time - minute	
├─ workMode	string	Not Required		Working mode	
├─ extraParam	object	Not Required			
├─ minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Not Required		Max soc value	
├─ importLimit	number	Not Required		Import Limit	
├─ exportLimit	number	Not Required		Export Limit	

Name	Type	Is it required	Default value	Note	Other information
├─ pvLimit	number	Not Required		PV Limit	
├─ reactivePower	number	Not Required		Reactive Power Limit	

Scheduler-V2

Get the time segment information

Basic information

Path /op/v2/device/scheduler/get

Method POST

Interface description

Get timing mode segment information

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ enable	string	Required		State of master switch (0:off 1:on)	
├─ groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ extraParam	object	Required			
├─ minSocOnGrid	number	Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Required		Max soc value	
├─ importLimit	number	Required		Import Limit	
├─ exportLimit	number	Required		Export Limit	

Name	Type	Is it required	Default value	Note	Other information
├─ pvLimit	number	Required		PV Limit	
├─ reactivePower	number	Required		Reactive Power Limit	
├─ properties	object	Required		Range for each field in the group time period	
├─ {field_name}	object	Required			
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	

Set the time segment information

Basic information

Path /op/v2/device/scheduler/enable

Method POST

Interface description

Set the time segment information of the timing mode

2025-12-25: Add the parameter extraParamDefaultValueSwitch, and make the parameters in extraParam optional (non-mandatory).

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
isDefault	boolean	Not Required	false	- false: Parameters not provided in `extraParam` remain unchanged. - true: Parameters not provided in `extraParam` are restored to system defaults.	
groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ extraParam	object	Not Required			
├─ minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Not Required		Max soc value	
├─ importLimit	number	Not Required		Import Limit	
├─ exportLimit	number	Not Required		Export Limit	
├─ pvLimit	number	Not Required		PV Limit	

Name	Type	Is it required	Default value	Note	Other information
⊢ reactivePower	number	Not Required		Reactive Power Limit	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ isDefault	boolean	Not Required	false	- false: Parameters not provided in `extraParam` remain unchanged. - true: Parameters not provided in `extraParam` are restored to system defaults.	
├─ groups	object []	Not Required			item Type: object
├─ enable	integer	Not Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Not Required		Start time - hour	
├─ startMinute	integer	Not Required		Start time - minute	
├─ endHour	integer	Not Required		Ending time - hour	
├─ endMinute	integer	Not Required		Ending time - minute	
├─ workMode	string	Not Required		Working mode	
├─ extraParam	object	Not Required			
├─ minSocOnGrid	number	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	number	Not Required		The FC/FDSoC is the SoC level set for the system when operating in Force Charge/Force Discharge Mode. Once the battery reaches this SoC, the system will automatically stop charging or discharging.	
├─ fdPwr	number	Not Required		The FC/FDPwr defines the maximum charging or discharging power allowed when the system is in Force Charge/Force Discharge Mode. The maximum AC input power the system can draw from the grid to charge the battery in Force Charge. The maximum AC output power the system is allowed to deliver during Forced Discharge.	
├─ maxSoc	number	Not Required		Max soc value	
├─ importLimit	number	Not Required		Import Limit	

Name	Type	Is it required	Default value	Note	Other information
├─ exportLimit	number	Not Required		Export Limit	
├─ pvLimit	number	Not Required		PV Limit	
├─ reactivePower	number	Not Required		Reactive Power Limit	

Scheduler-V1

Get the main switch status

Basic information

Path /op/v1/device/scheduler/get/flag

Method POST

Interface description

Obtain the timing mode switch status

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ support	boolean	Required		Whether the timing function is supported	
├─ enable	boolean	Required		Whether to enable the timing function	

Set the main switch status

Basic information

Path /op/v1/device/scheduler/set/flag

Method POST

Interface description

Set the timing mode switch

request body example:

```
{
    "deviceSN": "**************",
    "enable": 1
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
enable	integer	Required		Whether the switch is on 0:disable 1:enable	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ enable	integer	Not Required		Whether the switch is on 0:disable 1:enable	

Get the time segment information

Basic information

Path /op/v1/device/scheduler/get

Method POST

Interface description

Get timing mode segment information

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ enable	string	Required		State of master switch (0:off 1:on)	
├─ groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ minSocOnGrid	integer	Required		The minimum soc value of the offline battery	
├─ fdSoc	integer	Required		Discharge soc value	
├─ fdPwr	integer	Required		The maximum discharge power value	
├─ maxSoc	string	Required		Max soc value	
├─ properties	object	Required		Range for each field in the group time period	
├─ {field_name}	object	Required			
├─ unit	string	Required		unit	
├─ precision	number	Required		precision	Enumeration: 1,0.1,0.01
├─ range	object	Required			
├─ min	number	Required		minimum	
├─ max	number	Required		maximum	

Set the time segment information

Basic information

Path /op/v1/device/scheduler/enable

Method POST

Interface description

Set the time segment information of the timing mode

request body example:

```
{
  "deviceSN": "************",
  "groups": [
    {
      "enable": 1,
      "startHour": 0,
      "startMinute": 0,
      "endHour": 5,
      "endMinute": 0,
      "workMode": "SelfUse",
      "minSocOnGrid": 10,
      "fdSoc": 90,
      "fdPwr": 3000,
      "maxSoc": 100
    },
    {
      "enable": 1,
      "startHour": 6,
      "startMinute": 0,
      "endHour": 17,
      "endMinute": 0,
      "workMode": "Feedin",
      "minSocOnGrid": 10,
      "fdSoc": 90,
      "fdPwr": 3000,
      "maxSoc": 100
    }
  ]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour	
├─ startMinute	integer	Required		Start time - minute	
├─ endHour	integer	Required		Ending time - hour	
├─ endMinute	integer	Required		Ending time - minute	
├─ workMode	string	Required		Working mode	Enumeration: SelfUse,Feedin,Backup,ForceCharge,ForceDischarge
├─ minSocOnGrid	integer	Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	integer	Required		Discharge soc range value of fdsoc must be equal or greater than minSocOnGrid	
├─ fdPwr	number	Required		The maximum discharge power	
├─ maxSoc	integer	Not Required		The maximum soc	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ groups	object []	Not Required			item Type: object
├─ enable	integer	Not Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Not Required		Start time - hour	
├─ startMinute	integer	Not Required		Start time - minute	
├─ endHour	integer	Not Required		Ending time - hour	
├─ endMinute	integer	Not Required		Ending time - minute	
├─ workMode	string	Not Required		Working mode	
├─ minSocOnGrid	integer	Not Required		Battery discharge cutoff SOC in grid-connected state	
├─ fdSoc	integer	Not Required		Discharge soc range value of fdsoc must be equal or greater than minSocOnGrid	
├─ fdPwr	number	Not Required		The maximum discharge power	
├─ maxSoc	integer	Not Required		The maximum soc	

Scheduler

Get the main switch status (Deprecated)

Basic information

Path /op/v0/device/scheduler/get/flag

Method POST

Interface description

Obtain the timing mode switch status

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ support	integer	Required		Whether the timing function is supported(0:nonsupport 1:support) 0:nonsupport 1:support	
├─ enable	integer	Required		Whether to enable the timing function(0:disable 1:enable) 0:disable 1:enable	

Get the time segment information (Deprecated)

Basic information

Path /op/v0/device/scheduler/get

Method POST

Interface description

Get timing mode segment information

request body example:

```
{
    "deviceSN": "**************"
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		error number When the result is not equal to zero, the request fails	
msg	string	Required		Error message	
result	object	Required			
├─ enable	string	Required		State of master switch (0:off 1:on)	
├─ groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour(The value is between 0 and 23)	
├─ startMinute	integer	Required		Start time - minute(The value is between 0 and 59)	
├─ endHour	integer	Required		Ending time - hour(The value is between 0 and 23)	
├─ endMinute	integer	Required		Ending time - minute(The value is between 0 and 59)	
├─ workMode	string	Required		Working mode SelfUse Feedin Backup ForceCharge ForceDischarge	
├─ minSocOnGrid	integer	Required		socThe minimum soc value of the offline battery	
├─ fdSoc	integer	Required		Discharge soc value	
├─ fdPwr	integer	Required		The maximum discharge power value	

Set the main switch status (Deprecated)

Basic information

Path /op/v0/device/scheduler/set/flag

Method POST

Interface description

Set the timing mode switch

request body example:

```
{
    "deviceSN": "**************",
    "enable": 1
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
enable	integer	Required		Whether the switch is on 0:disable 1:enable	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ enable	integer	Not Required		Whether the switch is on 0:disable 1:enable	

Set the time segment information (Deprecated)

Basic information

Path /op/v0/device/scheduler/enable

Method POST

Interface description

Set the time segment information of the timing mode

request body example:

```
{
  "deviceSN": "************",
  "groups": [
    {
      "enable": 1,
      "startHour": 0,
      "startMinute": 0,
      "endHour": 5,
      "endMinute": 0,
      "workMode": "SelfUse",
      "minSocOnGrid": 10,
      "fdSoc": 90,
      "fdPwr": 3000
    },
    {
      "enable": 1,
      "startHour": 6,
      "startMinute": 0,
      "endHour": 17,
      "endMinute": 0,
      "workMode": "Feedin",
      "minSocOnGrid": 10,
      "fdSoc": 90,
      "fdPwr": 3000
    }
  ]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
deviceSN	string	Required		Serial number of device	
groups	object []	Required			item Type: object
├─ enable	integer	Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Required		Start time - hour(The value is between 0 and 23)	
├─ startMinute	integer	Required		Start time - minute(The value is between 0 and 59)	
├─ endHour	integer	Required		Ending time - hour(The value is between 0 and 23)	
├─ endMinute	integer	Required		Ending time - minute(The value is between 0 and 59)	
├─ workMode	string	Required		Working mode SelfUse Feedin Backup ForceCharge ForceDischarge	
├─ minSocOnGrid	integer	Required		The minimum soc range of the offline battery is between 10 and 100	
├─ fdSoc	integer	Required		Discharge soc range between 0 and 100,value of fdsoc must be equal or greater than minSocOnGrid value of fdsoc must be equal or greater than minSocOnGrid	
├─ fdPwr	number	Required		The maximum discharge power ranges between 0 and 6000	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ deviceSN	string	Not Required		Serial number of device	
├─ groups	object []	Not Required			item Type: object
├─ enable	integer	Not Required		Whether to enable this group setting(0:disable 1:enable)	
├─ startHour	integer	Not Required		Start time - hour(The value is between 0 and 23)	
├─ startMinute	integer	Not Required		Start time - minute(The value is between 0 and 59)	
├─ endHour	integer	Not Required		Ending time - hour(The value is between 0 and 23)	
├─ endMinute	integer	Not Required		Ending time - minute(The value is between 0 and 59)	
├─ workMode	string	Not Required		Working mode SelfUse Feedin Backup ForceCharge ForceDischarge	
├─ minSocOnGrid	integer	Not Required		The minimum soc range of the offline battery is between 10 and 100	
├─ fdSoc	integer	Not Required		Discharge soc range between 0 and 100,value of fdsoc must be equal or greater than minSocOnGrid value of fdsoc must be equal or greater than minSocOnGrid	
├─ fdPwr	number	Not Required		The maximum discharge power ranges between 0 and 6000	

Data Logger

Get Data Logger list

Basic information

Path /op/v0/module/list

Method POST

Interface description

Obtain the collector list of this account

request body example:

```
{
  "currentPage": 1,
  "pageSize": 10
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Maximum value: 1000 Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ moduleSN	string	Required		Serial number of Data Logger	
├─ stationID	string	Required		Id of power station	
├─ status	integer	Required		Status of collector	Enumeration: 1,2 Enumeration note: 1:online 2:offline
├─ signal	integer	Required		Signal strength	Maximum value: 100 Minimum value: 0
├─ webVersion	string	Required	--	Web Version	
├─ softVersion	string	Required	--	Soft Version	

Get Data Logger Details

Basic information

Path /op/v0/module/detail

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ moduleSN	string	Required		Serial number of Data Logger	
├─ stationID	string	Required		Id of power station	
├─ status	integer	Required		Status of collector	Enumeration: 1,2 Enumeration note: 1:online 2:offline
├─ signal	integer	Required		Signal strength	Maximum value: 100 Minimum value: 0
├─ webVersion	string	Required	--	Web Version	
├─ softVersion	string	Required	--	Soft Version	

Get Data Logger Signal

Basic information

Path /op/v0/module/getSignal

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ timestamp	integer	Required		Data recording timestamp	
├─ signal	integer	Required		Signal strength in dBm	

Modbus Commands

Basic information

Path /op/v0/module/modbus/commands

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			
timeout	number	Not Required		Timeout duration, in seconds. Default: 10s	
data	string	Required		The data content to be distributed, base64-encoded (including headers and trailers)	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
msg	string	Required			
result	object	Required			
├─ data	string	Required			

Get Data Logger Wifi Info

Basic information

Path /op/v0/module/getWifiInfo

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ ssid	string	Required		The name of the Wi-Fi network	
├─ ip	string	Required		The IP address	
├─ gateway	string	Required		The IP address of the network gateway	
├─ mask	string	Required		The subnet mask used to identify the network segment	

Get Data Logger Lan Info

Basic information

Path /op/v0/module/getLanInfo

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ ip	string	Required		The IP address	
├─ gateway	string	Required		The IP address of the network gateway	
├─ mask	string	Required		The subnet mask used to identify the network segment	

EMS

Get Ems list

Basic information

Path /op/v0/ems/list

Method POST

Interface description

Obtain the ems list of this account

request body example:

```
{
  "currentPage": 1,
  "pageSize": 10
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Maximum value: 1000 Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ emsID	string	Required		Id of Ems	
├─ emsSN	string	Required		Serial number of Ems	
├─ plantID	string	Required		Id of power station	
├─ productType	string	Required		product type	
├─ status	string	Required		Status of Ems	Enumeration: 0,1 Enumeration note: online offline

Get Ems real-time data

Basic information

Path /op/v0/ems/real/query

Method POST

Interface description

Get the latest real-time data of the inverter. The sn array is mandatory and returns only the result set belonging to the current user. If no variable is specified, all variables are queried

request body example:

```
{
  "sns": ["***************","***************","***************"],
  "variables": ["totalFeedinPower","totalLoadsPower"]
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
variables	string []	Required		Search variables item type: string	item Type: string
├─		Not Required		Variable name Available variables can be queried through the Get Variables interface	
sns	string []	Required		Serial Number of Ems Remark:Can transmit up to 50 Serial Number	item Type: string
├─		Not Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object []	Required			item Type: object
├─ emsSN	string	Required		Serial Number of Ems	
├─ time	string	Required		Time of Data Update, utc time	
├─ datas	object []	Required			item Type: object
├─ variable	string	Required		Variable name If the data is not found, it will not be returned	
├─ unit	string	Required		Unit	
├─ name	string	Required		Name in English	
├─ value	string	Required		Value	

Get Ems history data

Basic information

Path /op/v0/ems/history/query

Method POST

Interface description

Get the latest real-time data of the inverter. The sn array is mandatory and returns only the result set belonging to the current user. If no variable is specified, all variables are queried

request body example:

```
{
  "sn": "***************",
  "variables": ["totalFeedinPower","totalLoadsPower"],
  "begin": xxxxxx,
  "end": xxxxxx
}
```

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
variables	string []	Required		Search variables item type: string	item Type: string
├─		Not Required		Variable name Available variables can be queried through the Get Variables interface	
sn	string	Required		Serial Number of Ems	
begin	number	Not Required		The start timestamp for the query(milliseconds) The time range between begin and end must not exceed 24 hours.	
end	number	Not Required		The end timestamp for the query(milliseconds) The time range between begin and end must not exceed 24 hours.	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object []	Required			item Type: object
├─ emsSN	string	Required		Serial Number of Ems	
├─ datas	object []	Required			item Type: object
├─ variable	string	Required		Variable name If the data is not found, it will not be returned	
├─ unit	string	Required		Unit	
├─ name	string	Required		Name in English	
├─ data	object []	Required			item Type: object
├─ time	string	Required		Time of Data Update	
├─ value	number	Required		Value	

Get Ems System Setting

Basic information

Path /op/v0/ems/setting/system/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
emsID	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required		title	
├─ general	object	Required			
├─ drm	boolean	Required			
├─ rippleControl	boolean	Required			
├─ dataReportingCycle	integer	Required		200-60000 ms	
├─ parallelControl	object	Required			
├─ enable	boolean	Required			
├─ controlMode	string	Required			
├─ controlPercentage	integer	Required			
├─ dataReportingCycle	number	Required		0.1-10 s	
├─ powerLimit	object	Required			
├─ enable	boolean	Required			
├─ exportPowerLimit	integer	Required			
├─ importPowerLimit	integer	Required			
├─ feedinPowerLimit	integer	Required			

Set Ems System Setting

Basic information

Path /op/v0/ems/setting/system/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
emsID	string	Required			
general	object	Required			
├─ drm	boolean	Required			
├─ rippleControl	boolean	Required			
├─ dataReportingCycle	integer	Required		200-60000 ms	
parallelControl	object	Required			
├─ enable	boolean	Required			
├─ controlMode	string	Required			Enumeration: totalPowerControl,maximumSinglePowerControl Enumeration note:
├─ controlPercentage	integer	Required		0-100%	
├─ dataReportingCycle	number	Required		0.1-10 s	
powerLimit	object	Required			
├─ enable	boolean	Required			
├─ exportPowerLimit	integer	Required			
├─ importPowerLimit	integer	Required			
├─ feedinPowerLimit	integer	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ emsID	string	Not Required			
├─ general	object	Not Required			
├─ drm	boolean	Required			
├─ rippleControl	boolean	Required			
├─ dataReportingCycle	integer	Required		200-60000 ms	
├─ parallelControl	object	Not Required			
├─ enable	boolean	Required			
├─ controlMode	string	Required			Enumeration: totalPowerControl,maximumSinglePowerControl Enumeration note:
├─ controlPercentage	integer	Required		0-100%	
├─ dataReportingCycle	number	Required		0.1-10 s	
├─ powerLimit	object	Not Required			
├─ enable	boolean	Required			
├─ exportPowerLimit	integer	Required			
├─ importPowerLimit	integer	Required			
├─ feedinPowerLimit	integer	Required			

Get Ems Rate Setting

Basic information

Path /op/v0/ems/setting/rate/get

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
emsID	string	Required			
port	integer	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ dataBit	integer	Required		5-8	
├─ stopBit	number	Required		1 1.5 2	
├─ parityBit	integer	Required		0: 2: 1:	
├─ rate	integer	Required		9600-115200 RequiredRequired1200	

Set Ems Rate Setting

Basic information

Path /op/v0/ems/setting/rate/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
emsID	string	Required			
port	integer	Required			
rate	integer	Required		9600-115200 RequiredRequired1200	
dataBit	integer	Required		5-8	
stopBit	number	Required		1 1.5 2	
parityBit	integer	Required		0: 2: 1:	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ emsID	string	Not Required			
├─ port	integer	Not Required			
├─ rate	integer	Not Required		9600-115200 RequiredRequired1200	
├─ dataBit	integer	Not Required		5-8	
├─ stopBit	number	Not Required		1 1.5 2	
├─ parityBit	integer	Not Required		0: 2: 1:	

Set Ems Ac Output Control Setting

Basic information

Path /op/v0/ems/setting/acOutput/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		ems SN	
drmEnable	string	Required			Enumeration: disable,enable
rippleControl	string	Required			Enumeration: disable,enable
rippleControlPiecewise1	string	Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
rippleControlPiecewise2	string	Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
rippleControlPiecewise3	string	Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
rippleControlPiecewise4	string	Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		ems SN	
├─ drmEnable	string	Not Required			Enumeration: disable,enable
├─ rippleControl	string	Not Required			Enumeration: disable,enable
├─ rippleControlPiecewise1	string	Not Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
├─ rippleControlPiecewise2	string	Not Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
├─ rippleControlPiecewise3	string	Not Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	
├─ rippleControlPiecewise4	string	Not Required		The value must be a decimal between 0 and 100. It has no effect when rippleControl is set to false	

Get Ems Ac Output Control Setting

Basic information

Path /op/v0/ems/setting/acOutput/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ drmEnable	object	Required			
├─ value	string	Required			
├─ enumList	string []	Required		Possible values for value	item Type: string
├─		Not Required			
├─ rippleControl	object	Required			
├─ value	string	Required			
├─ enumList	string []	Required		Possible values for value	item Type: string
├─		Not Required			
├─ rippleControlPiecewise1	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ rippleControlPiecewise2	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ rippleControlPiecewise3	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			

Name	Type	Is it required	Default value	Note	Other information
├─ rippleControlPiecewise4	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			

Set Ems Power Limit Control Setting

Basic information

Path /op/v0/ems/setting/powerLimit/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		ems SN	
powerLimitEnable	string	Required			Enumeration: enable,disable
gridExportLimit	string	Required		Unit:W	
gridImportLimit	string	Required		Unit:W	
gridExportLimitForPeakShaving	string	Required		Unit:W	
batterySocLimitForPeakShaving	string	Required		Unit:%	
acOutputExportLimit	string	Required		Unit:W	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		ems SN	
├─ powerLimitEnable	string	Not Required			Enumeration: enable,disable
├─ gridExportLimit	string	Not Required		Unit:W	
├─ gridImportLimit	string	Not Required		Unit:W	
├─ gridExportLimitForPeakShaving	string	Not Required		Unit:W	
├─ batterySocLimitForPeakShaving	string	Not Required		Unit:%	
├─ acOutputExportLimit	string	Not Required		Unit:W	

Get Ems Power Limit Control Setting

Basic information

Path /op/v0/ems/setting/powerLimit/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ powerLimitEnable	object	Required			
├─ value	string	Required			
├─ enumList	string []	Required		Possible values for value	item Type: string
├─		Not Required			
├─ gridExportLimit	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ gridImportLimit	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ gridExportLimitForPeakShaving	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ batterySocLimitForPeakShaving	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	

Name	Type	Is it required	Default value	Note	Other information
├─ min	number	Required			
├─ max	number	Required			
├─ acOutputExportLimit	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			

Set Ems More Setting

Basic information

Path /op/v0/ems/setting/more/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		ems SN	
controlMode	string	Required			Enumeration: singlePhase,threePhase
controlCycle	string	Required			
gridExportControlDelta	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		ems SN	
├─ controlMode	string	Not Required			Enumeration: singlePhase,threePhase
├─ controlCycle	string	Not Required			
├─ gridExportControlDelta	string	Not Required			

Get Ems More Setting

Basic information

Path /op/v0/ems/setting/more/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ controlMode	object	Required			
├─ value	string	Required			
├─ enumList	string []	Required		Possible values for value	item Type: string
├─		Not Required			
├─ controlCycle	object	Required		Possible values for value	
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ gridExportControlDelta	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			

Set Ems Gen Setting

Basic information

Path /op/v0/ems/setting/gen/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		ems sn	
genEnable	string	Required			Enumeration: disable,enable
startSoc	string	Required			
stopSoc	string	Required			
genCharge	string	Required			
judgeTime	string	Required			
minimumResetTime	string	Required			
meterCompensation	string	Not Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		ems sn	
├─ genEnable	string	Not Required			Enumeration: disable,enable
├─ startSoc	string	Not Required			
├─ stopSoc	string	Not Required			
├─ genCharge	string	Not Required			
├─ judgeTime	string	Not Required			
├─ minimumResetTime	string	Not Required			
├─ meterCompensation	string	Not Required			

Get Ems Gen Setting

Basic information

Path /op/v0/ems/setting/gen/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Query

Parameter name	Is it required	Example	Note
sn	Required		

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ genEnable	object	Required			
├─ value	string	Required			
├─ enumList	string []	Required		Possible values for value	item Type: string
├─		Not Required			
├─ startSoc	object	Required		Possible values for value	
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ stopSoc	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ genCharge	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ judgeTime	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	

Name	Type	Is it required	Default value	Note	Other information
├─ min	number	Required			
├─ max	number	Required			
├─ minimumResetTime	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required		Valid range	
├─ min	number	Required			
├─ max	number	Required			
├─ meterCompensation	object	Not Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	string	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			

Meter

Get meter list

Basic information

Path /op/v0/gw/list

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ sn	string	Required		Serial number of the meter	
├─ stationID	string	Required		Id of power station	
├─ stationName	string	Required		Name of power station	
├─ status	integer	Required		Status of meter	Enumeration: 1,2,3 Enumeration note: 1:online 2:fault 3:offline

Get the device settings item

Basic information

Path /op/v0/gw/setting/get

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial number of Inverter	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
result	object	Required			
├─ mode	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ feedinPower	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ outputMaxCurrent	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ inputMaxCurrent	object	Required			
├─ value	string	Required			

Name	Type	Is it required	Default value	Note	Other information
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ overVoltageMaxValue	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ underVoltageMinValue	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ sysOnOffPowerValue	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			

Name	Type	Is it required	Default value	Note	Other information
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			
├─ deviceQuantity	object	Required			
├─ value	string	Required			
├─ unit	string	Required			
├─ precision	number	Required			
├─ range	object	Required			
├─ min	number	Required			
├─ max	number	Required			
├─ enumList	string []	Required			item Type: string
├─		Not Required			

Set the device settings item

Basic information

Path /op/v0/gw/setting/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			
mode	string	Required			
feedinPower	string	Not Required		Only available in non-normal mode	
outputMaxCurrent	string	Not Required		Only available in G100mode	
inputMaxCurrent	string	Not Required		Only available in G100mode	
overVoltageMaxValue	string	Not Required		Only available in G100mode	
underVoltageMinValue	string	Not Required		Only available in G100mode	
sysOnOffPowerValue	string	Not Required		Only available in G100mode	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required			
├─ mode	string	Not Required			
├─ feedinPower	string	Not Required		Only available in non-normal mode	
├─ outputMaxCurrent	string	Not Required		Only available in G100mode	
├─ inputMaxCurrent	string	Not Required		Only available in G100mode	
├─ overVoltageMaxValue	string	Not Required		Only available in G100mode	
├─ underVoltageMinValue	string	Not Required		Only available in G100mode	
├─ sysOnOffPowerValue	string	Not Required		Only available in G100mode	

Heat

Heat Pump Register

Basic information

Path /op/v0/heat/register

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Outdoor Unit of Heat Pump SN	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
msg	string	Required			
result	object	Required		title	
├─ sn	string	Not Required		Outdoor Unit of Heat Pump SN	

Heat Pump Register List

Basic information

Path /op/v0/register/heat/list

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Minimum value: 10
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ heatSN	string	Required		Serial number of the Heat	
├─ moduleSN	string	Required		Module Serial number	
├─ registerStatus	string	Required		Register Status	
├─ runningStatus	integer	Required		Status of Heat(1 online;2 error;3 offline))	Enumeration: 1,2,3 Enumeration note: 1:online 2:fault 3:offline
├─ masterVersion	string	Required		version	
├─ deviceType	string	Required		deviceType of The Heat	

Heat Pump Register Status Change

Basic information

Path /op/v0/heat/register/status/change

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Outdoor Unit of Heat Pump SN	
status	string	Required		pending approved revoked	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ sn	string	Not Required		Outdoor Unit of Heat Pump SN	
├─ status	string	Not Required		pending approved revoked	

dhwControlsGet

Basic information

Path /op/v0/heat/dhwControls/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
moduleSn	Required		

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result	object	Not Required			
├─ enable	boolean	Not Required		Master switch to enable or disable the entire DHW function.	
├─ timer	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of timer entries. A maximum of 6 time periods are supported. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ dhwTemp	number	Not Required		Target temperature for domestic hot water (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ bivalencePoint	object	Not Required			
├─ electricHeatingWaterEnvironmentalTemp	number	Not Required		Ambient temperature threshold (°C) for the main heat source (e.g., heat pump). If the ambient temperature falls below this value, the auxiliary heater may be triggered. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ hotWaterElectricHeatingAmbientTemp	number	Not Required		Ambient temperature threshold (°C) specifically for DHW auxiliary electric heating. Determines if the dedicated DHW electric heater should be enabled. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ auxiliaryHeaterSettings	object	Not Required			
├─ electricHeatingControl	boolean	Not Required		Master control switch for the DHW auxiliary electric heater.	
├─ hotWaterElectricHeatingEnable	boolean	Not Required		Function enable switch for the DHW auxiliary electric heater.	

dhwControlsSet

Basic information

Path /op/v0/heat/dhwControls/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
moduleSn	string	Required			
data	object	Required			
├─ enable	boolean	Not Required		Master switch to enable or disable the entire DHW function.	
├─ timer	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of timer entries. A maximum of 6 time periods are supported. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ dhwTemp	number	Not Required		Target temperature for domestic hot water (°C). Note: The device accepts a precision of 0.1°C. Values will be	

Name	Type	Is it required	Default value	Note	Other information
				rounded (e.g., 0.06 becomes 0.1).	
├─ bivalencePoint	object	Not Required			
├─ electricHeatingWaterEnvironmentalTemp	number	Not Required		Ambient temperature threshold (°C) for the main heat source (e.g., heat pump). If the ambient temperature falls below this value, the auxiliary heater may be triggered. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ hotWaterElectricHeatingAmbientTemp	number	Not Required		Ambient temperature threshold (°C) specifically for DHW auxiliary electric heating. Determines if the dedicated DHW electric heater should be enabled. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ auxiliaryHeaterSettings	object	Not Required			
├─ electricHeatingControl	boolean	Not Required		Master control switch for the DHW auxiliary electric heater.	
├─ hotWaterElectricHeatingEnable	boolean	Not Required		Function enable switch for the DHW auxiliary electric heater.	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result		Not Required			

genericControlsGet

Basic information

Path /op/v0/heat/genericControls/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
moduleSn	Required		

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result	object	Not Required			
├─ holiday	object	Not Required			
├─ away	object	Not Required			
├─ enable	boolean	Not Required		Enable or disable the 'Holiday Away' mode.	
├─ startYear	integer	Not Required		Start year for 'Holiday Away' mode.	
├─ startMonth	integer	Not Required		Start month for 'Holiday Away' mode.	Maximum value: 12 Minimum value: 1
├─ startDay	integer	Not Required		Start day for 'Holiday Away' mode.	Maximum value: 31 Minimum value: 1
├─ endYear	integer	Not Required		End year for 'Holiday Away' mode.	
├─ endMonth	integer	Not Required		End month for 'Holiday Away' mode.	Maximum value: 12 Minimum value: 1
├─ endDay	integer	Not Required		End day for 'Holiday Away' mode.	Maximum value: 31 Minimum value: 1
├─ heatingModeEnable	boolean	Not Required		Enable or disable heating function during 'Holiday Away' mode.	
├─ heatingTemp	number	Not Required		Target heating temperature (°C) during 'Holiday Away' mode. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ dhwModeEnable	boolean	Not Required		Enable or disable Domestic Hot Water (DHW) function during 'Holiday Away' mode.	
├─ dhwTemp	number	Not Required		Target DHW temperature (°C) during 'Holiday Away' mode. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ disinfect	boolean	Not Required		Enable or disable the disinfection function during 'Holiday Away' mode.	
├─ home	object	Not Required			
├─ enable	boolean	Not Required		Enable or disable the 'Holiday Home' mode.	
├─ startYear	integer	Not Required		Start year for 'Holiday Home' mode.	
├─ startMonth	integer	Not Required		Start month for 'Holiday Home' mode.	Maximum value: 12 Minimum value: 1
├─ startDay	integer	Not Required		Start day for 'Holiday Home' mode.	Maximum value: 31 Minimum value: 1
├─ endYear	integer	Not Required		End year for 'Holiday Home' mode.	
├─ endMonth	integer	Not Required		End month for 'Holiday Home' mode.	Maximum value: 12 Minimum value: 1
├─ endDay	integer	Not Required		End day for 'Holiday Home' mode.	Maximum value: 31 Minimum value: 1
├─ zone1	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object

Name	Type	Is it required	Default value	Note	Other information
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ zone2	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone3	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object[]	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ dhw	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	

genericControlsSet

Basic information

Path /op/v0/heat/genericControls/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
moduleSn	string	Required			
data	object	Required			
├─ holiday	object	Not Required			
├─ away	object	Not Required			
├─ enable	boolean	Not Required		Enable or disable the 'Holiday Away' mode.	
├─ startYear	integer	Not Required		Start year for 'Holiday Away' mode.	
├─ startMonth	integer	Not Required		Start month for 'Holiday Away' mode.	Maximum value: 12 Minimum value: 1
├─ startDay	integer	Not Required		Start day for 'Holiday Away' mode.	Maximum value: 31 Minimum value: 1
├─ endYear	integer	Not Required		End year for 'Holiday Away' mode.	
├─ endMonth	integer	Not Required		End month for 'Holiday Away' mode.	Maximum value: 12 Minimum value: 1
├─ endDay	integer	Not Required		End day for 'Holiday Away' mode.	Maximum value: 31 Minimum value: 1
├─ heatingModeEnable	boolean	Not Required		Enable or disable heating function during 'Holiday Away' mode.	
├─ heatingTemp	number	Not Required		Target heating temperature (°C) during 'Holiday Away' mode. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ dhwModeEnable	boolean	Not Required		Enable or disable Domestic Hot Water (DHW) function during 'Holiday Away' mode.	
├─ dhwTemp	number	Not Required		Target DHW temperature (°C) during 'Holiday Away' mode. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g.,	

Name	Type	Is it required	Default value	Note	Other information
				0.06 becomes 0.1).	
├─ disinfect	boolean	Not Required		Enable or disable the disinfection function during 'Holiday Away' mode.	
├─ home	object	Not Required			
├─ enable	boolean	Not Required		Enable or disable the 'Holiday Home' mode.	
├─ startYear	integer	Not Required		Start year for 'Holiday Home' mode.	
├─ startMonth	integer	Not Required		Start month for 'Holiday Home' mode.	Maximum value: 12 Minimum value: 1
├─ startDay	integer	Not Required		Start day for 'Holiday Home' mode.	Maximum value: 31 Minimum value: 1
├─ endYear	integer	Not Required		End year for 'Holiday Home' mode.	
├─ endMonth	integer	Not Required		End month for 'Holiday Home' mode.	Maximum value: 12 Minimum value: 1
├─ endDay	integer	Not Required		End day for 'Holiday Home' mode.	Maximum value: 31 Minimum value: 1
├─ zone1	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object[]	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone2	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object[]	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ zone3	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ dhw	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result		Not Required			

heatingCircuitsControlsGet

Basic information

Path /op/v0/heat/heatingCircuitsControls/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
moduleSn	Required		

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result	object	Not Required			
├─ circuitActivation	object	Not Required			
├─ zone1Support	boolean	Not Required		Indicates if Zone 1 supports independent circuit control. Must be set to true before zone1Enable can be applied.	
├─ zone2Support	boolean	Not Required		Indicates if Zone 2 supports independent circuit control. Must be set to true before zone2Enable can be applied.	
├─ zone3Support	boolean	Not Required		Indicates if Zone 3 supports independent circuit control. Must be set to true before zone3Enable can be applied.	
├─ circuitType	object	Not Required			
├─ zone1Enable	boolean	Not Required		Indicates if Zone 1's circuit is enabled. zone1Support must be true for this field to take effect.	
├─ zone2Enable	boolean	Not Required		Indicates if Zone 2's circuit is enabled. zone2Support must be true for this field to take effect.	
├─ zone3Enable	boolean	Not Required		Indicates if Zone 3's circuit is enabled. zone3Support must be true for this field to take effect.	
├─ zone1	object	Not Required			
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1

Name	Type	Is it required	Default value	Note	Other information
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ mediumEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ zone2	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ zone3	object	Not Required			
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

heatingCircuitsControlsSet

Basic information

Path /op/v0/heat/heatingCircuitsControls/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
moduleSn	string	Required			
data	object	Required			
├─ circuitActivation	object	Not Required			
├─ zone1Support	boolean	Not Required		Indicates if Zone 1 supports independent circuit control. Must be set to true before zone1Enable can be applied.	
├─ zone2Support	boolean	Not Required		Indicates if Zone 2 supports independent circuit control. Must be set to true before zone2Enable can be applied.	
├─ zone3Support	boolean	Not Required		Indicates if Zone 3 supports independent circuit control. Must be set to true before zone3Enable can be applied.	
├─ circuitType	object	Not Required			
├─ zone1Enable	boolean	Not Required		Indicates if Zone 1's circuit is enabled. zone1Support must be true for this field to take effect.	
├─ zone2Enable	boolean	Not Required		Indicates if Zone 2's circuit is enabled. zone2Support must be true for this field to take effect.	
├─ zone3Enable	boolean	Not Required		Indicates if Zone 3's circuit is enabled. zone3Support must be true for this field to take effect.	
├─ zone1	object	Not Required			
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumEconomic	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ zone2	object	Not Required			
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	

Name	Type	Is it required	Default value	Note	Other information
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ zone3	object	Not Required			
├─ enable	boolean	Not Required		Switch to enable or disable the heating curve function for this zone.	
├─ temperatureLevel	integer	Not Required		Water Temperature Level: 1 - High Water Temperature, 2 - Medium Water Temperature, 3 - Low Water Temperature.	Maximum value: 3 Minimum value: 1
├─ temperatureMode	integer	Not Required		Temperature Mode: 1 - Warmth, 2 - Comfort, 3 - Economical.	Maximum value: 3 Minimum value: 1
├─ heatingCureOffsetValue	number	Not Required		Global offset value for the heating curve (used to fine-tune the outlet water temperature). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ highEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ mediumEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ lowComfort	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowWarm	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ lowEconomic	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Name	Type	Is it required	Default value	Note	Other information
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ cooling	object	Not Required			
├─ outdoorTemperatureLower	number	Not Required		Lower limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ outdoorTemperatureUpper	number	Not Required		Upper limit of outdoor temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureLower	number	Not Required		Lower limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ waterTemperatureUpper	number	Not Required		Upper limit of supply water temperature (°C). Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result		Not Required			

heatingControlsGet

Basic information

Path /op/v0/heat/heatingControls/get

Method GET

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Query

Parameter name	Is it required	Example	Note
moduleSn	Required		

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result	object	Not Required			
├─ workMode	integer	Not Required		System operating mode setting. 1 - Cooling, 2 - Heating, 3 - Auto, 4 - OFF AC	Maximum value: 4 Minimum value: 1
├─ zone1Daily	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone2Daily	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object[]	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone3Daily	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	

Name	Type	Is it required	Default value	Note	Other information
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ dhwDaily	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ zone1Weekly	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ mon	boolean	Not Required		Apply this timer to Monday.	
├─ tue	boolean	Not Required		Apply this timer to Tuesday.	
├─ wed	boolean	Not Required		Apply this timer to Wednesday.	
├─ thur	boolean	Not Required		Apply this timer to Thursday.	

Name	Type	Is it required	Default value	Note	Other information
├── fri	boolean	Not Required		Apply this timer to Friday.	
├── sat	boolean	Not Required		Apply this timer to Saturday.	
├── sun	boolean	Not Required		Apply this timer to Sunday.	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone2Weekly	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├── timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ mon	boolean	Not Required		Apply this timer to Monday.	
├─ tue	boolean	Not Required		Apply this timer to Tuesday.	
├─ wed	boolean	Not Required		Apply this timer to Wednesday.	
├─ thur	boolean	Not Required		Apply this timer to Thursday.	
├─ fri	boolean	Not Required		Apply this timer to Friday.	
├─ sat	boolean	Not Required		Apply this timer to Saturday.	
├─ sun	boolean	Not Required		Apply this timer to Sunday.	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ zone3Weekly	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 0
├— startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├— endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├— endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├— targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├— mon	boolean	Not Required		Apply this timer to Monday.	
├— tue	boolean	Not Required		Apply this timer to Tuesday.	
├— wed	boolean	Not Required		Apply this timer to Wednesday.	
├— thur	boolean	Not Required		Apply this timer to Thursday.	
├— fri	boolean	Not Required		Apply this timer to Friday.	
├— sat	boolean	Not Required		Apply this timer to Saturday.	
├— sun	boolean	Not Required		Apply this timer to Sunday.	

Name	Type	Is it required	Default value	Note	Other information
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── dhwWeekly	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├── timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── mon	boolean	Not Required		Apply this timer to Monday.	

Name	Type	Is it required	Default value	Note	Other information
├─ tue	boolean	Not Required		Apply this timer to Tuesday.	
├─ wed	boolean	Not Required		Apply this timer to Wednesday.	
├─ thur	boolean	Not Required		Apply this timer to Thursday.	
├─ fri	boolean	Not Required		Apply this timer to Friday.	
├─ sat	boolean	Not Required		Apply this timer to Saturday.	
├─ sun	boolean	Not Required		Apply this timer to Sunday.	
├─ bivalencePoint	object	Not Required			
├─ electricHeatingWaterEnvironmentalTemp	number	Not Required		Ambient temperature threshold (°C) for electric heating of supply water. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ airConditioningElectricHeatingAmbientTemp	number	Not Required		Ambient temperature threshold (°C) for air conditioning electric heating. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ auxiliaryHeaterSettings	object	Not Required			
├─ electricHeatingControl	boolean	Not Required		Master switch for electric heating control.	
├─ airConditioningElectricHeatingEnable	boolean	Not Required		Function enable switch for air conditioning electric heating.	

heatingControlsSet

Basic information

Path /op/v0/heat/heatingControls/set

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
moduleSn	string	Required			
data	object	Required			
├─ workMode	integer	Not Required		System operating mode setting. 1 - Cooling, 2 - Heating, 3 - Auto, 4 - OFF AC	Maximum value: 4 Minimum value: 1
├─ zone1Daily	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone2Daily	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	

Name	Type	Is it required	Default value	Note	Other information
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── zone3Daily	object	Not Required			
├── timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├── timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2

Name	Type	Is it required	Default value	Note	Other information
					Minimum value: 1
├─ dhwDaily	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the daily timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ zone1Weekly	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	

Name	Type	Is it required	Default value	Note	Other information
├── timers	object[]	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├── enable	boolean	Not Required		Whether this timer item is enabled	
├── startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── mon	boolean	Not Required		Apply this timer to Monday.	
├── tue	boolean	Not Required		Apply this timer to Tuesday.	
├── wed	boolean	Not Required		Apply this timer to Wednesday.	
├── thur	boolean	Not Required		Apply this timer to Thursday.	
├── fri	boolean	Not Required		Apply this timer to Friday.	

Name	Type	Is it required	Default value	Note	Other information
├─ sat	boolean	Not Required		Apply this timer to Saturday.	
├─ sun	boolean	Not Required		Apply this timer to Sunday.	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ zone2Weekly	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ mon	boolean	Not Required		Apply this timer to Monday.	
├─ tue	boolean	Not Required		Apply this timer to Tuesday.	
├─ wed	boolean	Not Required		Apply this timer to Wednesday.	
├─ thur	boolean	Not Required		Apply this timer to Thursday.	
├─ fri	boolean	Not Required		Apply this timer to Friday.	
├─ sat	boolean	Not Required		Apply this timer to Saturday.	
├─ sun	boolean	Not Required		Apply this timer to Sunday.	
├─ workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├─ zone3Weekly	object	Not Required			
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0

Name	Type	Is it required	Default value	Note	Other information
├── startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├── endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├── targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├── mon	boolean	Not Required		Apply this timer to Monday.	
├── tue	boolean	Not Required		Apply this timer to Tuesday.	
├── wed	boolean	Not Required		Apply this timer to Wednesday.	
├── thur	boolean	Not Required		Apply this timer to Thursday.	
├── fri	boolean	Not Required		Apply this timer to Friday.	
├── sat	boolean	Not Required		Apply this timer to Saturday.	
├── sun	boolean	Not Required		Apply this timer to Sunday.	
├── workMode	integer	Not Required		Operating mode: 1-Cooling; 2-Heating.	Maximum value: 2 Minimum value: 1
├── dhwWeekly	object	Not Required			

Name	Type	Is it required	Default value	Note	Other information
├─ timerEnable	boolean	Not Required		Master switch to enable or disable the weekly timer.	
├─ timers	object []	Not Required		List of specific timer entries. Max 6 entries allowed. Note: Date fields must be either all present or all absent - no partial dates.	: 6 item Type: object
├─ enable	boolean	Not Required		Whether this timer item is enabled	
├─ startHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ startMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ endHour	integer	Not Required			Maximum value: 23 Minimum value: 0
├─ endMinute	integer	Not Required			Maximum value: 59 Minimum value: 0
├─ targetTemp	number	Not Required		Target temperature in degrees Celsius, with at most one decimal place	
├─ mon	boolean	Not Required		Apply this timer to Monday.	
├─ tue	boolean	Not Required		Apply this timer to Tuesday.	
├─ wed	boolean	Not Required		Apply this timer to Wednesday.	
├─ thur	boolean	Not Required		Apply this timer to Thursday.	

Name	Type	Is it required	Default value	Note	Other information
├─ fri	boolean	Not Required		Apply this timer to Friday.	
├─ sat	boolean	Not Required		Apply this timer to Saturday.	
├─ sun	boolean	Not Required		Apply this timer to Sunday.	
├─ bivalencePoint	object	Not Required			
├─ electricHeatingWaterEnvironmentalTemp	number	Not Required		Ambient temperature threshold (°C) for electric heating of supply water. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ airConditioningElectricHeatingAmbientTemp	number	Not Required		Ambient temperature threshold (°C) for air conditioning electric heating. Note: The device accepts a precision of 0.1°C. Values will be rounded (e.g., 0.06 becomes 0.1).	
├─ auxiliaryHeaterSettings	object	Not Required			
├─ electricHeatingControl	boolean	Not Required		Master switch for electric heating control.	
├─ airConditioningElectricHeatingEnable	boolean	Not Required		Function enable switch for air conditioning electric heating.	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Not Required		Error number When the result is not equal to zero, the request fails	
msg	string	Not Required		Error message	
result		Not Required			

GMAX

Get gmax list

Basic information

Path /op/v0/gmax/list

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ currentPage	integer	Required		Current page number	
├─ pageSize	integer	Required		The amount of data presented per page	
├─ total	integer	Required		Data volume total	
├─ data	object []	Required			item Type: object
├─ gmaxID	string	Required		Device ID	
├─ gmaxSN	string	Required		Device SN	
├─ sn	string	Required		Complete machine SN	
├─ pcsSN	string	Required		PCS SN	
├─ bmsSN	string	Required		BMS SN	
├─ packSN	string	Required		PACK SN	
├─ plantID	string	Required		plant ID	
├─ communication	integer	Required		the status of device, 1online 2fault 3offline	

Get history data

Basic information

Path /op/v0/gmax/history/query

Method POST

Interface description

Obtain the historical data of the inverter, Only the result set belonging to the current user sn is returned. Obtain the historical data of the last one days without specifying the time,and the time span must be less than or equal to 24 hours

request body example:

```
{
  "sns": "*****",
  "begin": 1703548800000,
  "end": 1703635200000
}
```

response body example:


```
{
  "errno": 0,
  "msg": "Operation successful",
  "result": [
    {
      "datas": [
        {
          "soc": 1.0,
          "chargingPower": "0.0",
          "dischargingPower": "0.0",
          "time": "2025-11-25 17:58:16 CST+0800",
          "soh": 100.0,
          "status": "3.0"
        },
        {
          "soc": 1.0,
          "chargingPower": "0.0",
          "dischargingPower": "0.0",
          "time": "2025-11-25 17:59:16 CST+0800",
          "soh": 100.0,
          "status": "3.0"
        }
      ]
    }
  ]
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial Number	
begin	string	Not Required		Start timestamp(milliseconds)	
end	string	Not Required		End timestamp(milliseconds)	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
msg	string	Required			
result	object []	Required			item Type: object
├─ datas	object []	Required			item Type: object
├─ time	string	Required			
├─ soc	number	Required		unit:%	
├─ soh	number	Required		unit:%	
├─ status	string	Required		0Shutdown, 1Charging, 2Discharge, 3Standby, 4Fault	
├─ chargingPower	string	Required		unit:kw	
├─ dischargingPower	string	Required		unit:kw	

Get gmax real data

Basic information

Path /op/v0/gmax/real/query

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required			

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ chargingPower	string	Required		unit:kw	
├─ dischargingPower	string	Required		unit:kw	
├─ soc	number	Required		unit:%	
├─ soh	number	Required		unit:%	
├─ status	string	Required		0Shutdown, 1Charging, 2Discharge, 3Standby, 4Fault	

Get peak and valley arbitrage

Basic information

Path /op/v0/gmax/peakValleyPower/get

Method POST

Interface description

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial Number of GMAX	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Not Required			
msg	string	Not Required			
result	object	Not Required			
├─ datas	object []	Required		The maximum length is 8	item Type: object
├─ peakPower	number	Required		Peak power KW ,The charging power is negative and the discharging power is positive.	
├─ enable	boolean	Required		true:open , false:close	
├─ peakEndTime	number	Required		Peak end time(seconds)	
├─ peakWeek	number	Required		1: valid on Monday, 2:valid on Tuesday, 3:valid on Wednesday, 4:valid on Thursday, 5:valid on Friday, 6:valid on Saturday, 7valid on Sunday 7, The number of peak-valley arbitrage periods is 8, and 8 fields need to be set	
├─ peakStartTime	number	Required		Peak start time(seconds)	
├─ peakStopSoc	number	Required		The range is 0 to 100	
├─ peakWorkMode	number	Required		0: Effective by time period; 1: Valid by month; 2: Valid by year -1: Invalid value The number of periods for peak-valley arbitrage is 8, and 8 fields need to be set	
├─ peakEnable	boolean	Required		true:open , false:close	
├─ sn	string	Required		Serial Number	
├─ timestamp	number	Required			

Set peak and valley arbitrage

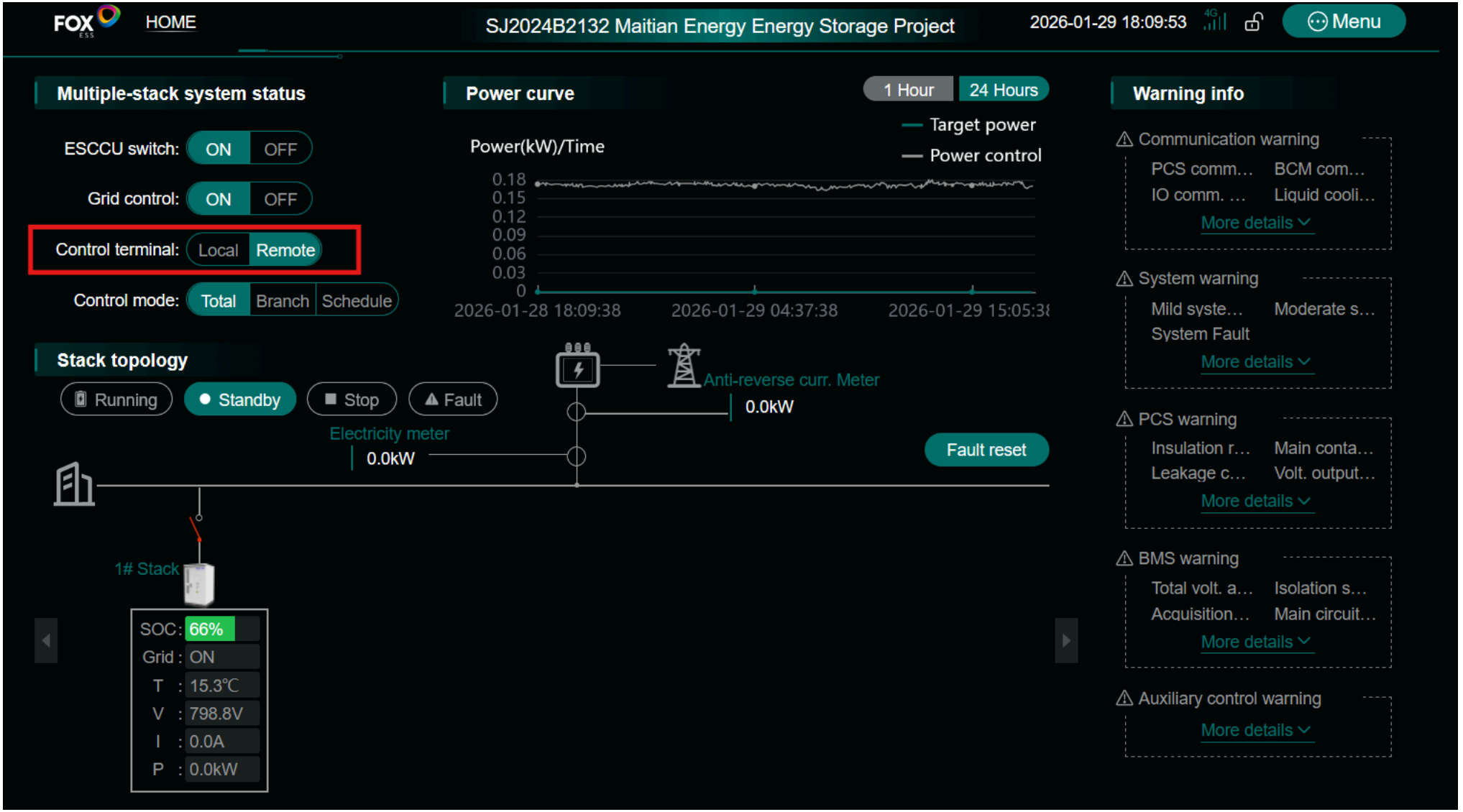
Basic information

Path /op/v0/gmax/peakValleyPower/set

Method POST

Interface description

Please ensure the control source is set to remote



Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required		Language

Body

Name	Type	Is it required	Default value	Note	Other information
sn	string	Required		Serial Number	
peakEnable	boolean	Required		true:open , false:close	
datas	object []	Required			: 1 : 8 item Type: object
├─ enable	boolean	Required		true:open , false:close	
├─ peakWorkMode	number	Required		0: Effective by time period; 1: Valid by month; 2: Valid by year -1: Invalid value The number of periods for peak-valley arbitrage is 8, and 8 fields need to be set	
├─ peakWeek	number	Required		1: valid on Monday, 2:valid on Tuesday, 3:valid on Wednesday, 4:valid on Thursday, 5:valid on Friday, 6:valid on Saturday, 7valid on Sunday 7, The number of peak-valley arbitrage periods is 8, and 8 fields need to be set	
├─ peakStopSoc	number	Required		The range is 0 to 100	
├─ peakPower	number	Required		Peak power KW ,The charging power is negative and the discharging power is positive.	
├─ peakStartTime	number	Required		Peak start time as Unix timestamp (seconds). Must be < peakEndTime, and its local clock time (HH:mm:ss) in the device's timezone must not be later than that of peakEndTime (e.g., 12:00 → 08:00 is invalid).	
├─ peakEndTime	number	Required		Peak end time as Unix timestamp (seconds). Must be > peakStartTime	

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
msg	string	Required			
result	string	Required			

Platform interface

Get the number of public interface accesses

Basic information

Path /op/v0/user/getAccessCount

Method GET

Interface description

Obtain the number of accessible interfaces under the account based on the open api.

Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information
currentPage	integer	Required	1	Current page number	Minimum value: 1
pageSize	integer	Required	10	The amount of data presented per page	Maximum value: 1000 Minimum value: 10

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	integer	Required		Error number When the result is not equal to zero, the request fails	
result	object	Required			
├─ total	string	Required		Number of total times that the user can access the open interface	
├─ remaining	string	Required		Number of remaining times that the user can access the open interface	

Installer device count

Basic information

Path /op/v0/device/installer/count

Method POST

Interface description

response body example:


```
{
  "errno": 0,
  "msg": "Operation successful",
  "result": null
}
```


Request parameter

Headers

Parameter name	Parameter value	Is it required	Example	Note
Content-Type	application/json	Required		
token		Required		Generate apikey from the API management function of the platform
signature		Required		Signature rule: Encrypt the string url + "\r\n" + token + "\r\n" + timestamp with md5
timestamp		Required		Current timestamp
lang		Required	en	Language

Body

Name	Type	Is it required	Default value	Note	Other information

Response Data

Name	Type	Is it required	Default value	Note	Other information
errno	number	Required			
msg	string	Required			
result	object []	Required			item Type: object
├─ year	number	Required			
├─ deviceCountryCounts	object []	Required			item Type: object
├─ country	string	Required			
├─ deviceCount	number	Required			